



MedEvo: Medical Reasoning Evolution through Knowledge Navigation

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Abstract

Medical multimodal reasoning models are typically trained on **teacher-generated rationales**, discarding the valid reasoning in students' own failed trajectories, while general-domain correction methods offer little guidance for errors in image reading, anatomical context, or differential diagnosis. We propose **MedEvo**, which uses **medical knowledge**, namely a tag taxonomy and error-type diagnosis, to navigate both **trajectory repair and data selection**. For each failed rollout, a **gold-blind auditor** locates the first consequential error and provides a minimal, answer-free hint; the same student **retains the valid prefix and regenerates the erroneous suffix**. After each round, MedEvo identifies capability gaps across medical tags on a **held-out feedback set** and prioritizes related questions for new rollouts and repairs. With **SFT alone**, MedEvo raises Qwen3.5-9B mean accuracy across seven benchmarks from **65.57% to 69.12%** (47.40% to **54.50%** on MedXpertQA-MM), surpassing teacher-distillation SFT (66.93%). It also raises Qwen3.5-35B and Qwen3.5-397B mean accuracy from 67.91% to **73.04%** and from 71.83% to **74.95%**, respectively. Our analyses show that failed trajectories contain reasoning worth preserving, that **repair success does not imply learning value**, and that **independent feedback is key to sustained improvement**.

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Code: <https://github.com/pork-pay/MedEvo>

Project: <https://pork-pay.github.io/MedEvo/>

1 Introduction

Medical multimodal reasoning requires connecting visual findings with anatomical context, disease knowledge, and clinical decisions, and expert-level questions remain challenging for current medical vision-language models [2, 3, 4]. A model may correctly recognize several findings yet reach a wrong answer after misreading one image feature, overlooking a clinical clue, or making an unsupported differential diagnosis. Such **failures are often local**: the reasoning before the mistake can remain valid, so preserving valid steps and recovering from the point of failure offers a direct way to improve medical reasoning [5, 6].

Existing approaches typically train students on teacher-generated rationales or on answer-correct student rollouts [7, 8, 9]. Discarding a failed trajectory loses its valid reasoning, while replacing it with a complete teacher solution gives the student a path it did not produce. Methods that learn from failures, such as mistake-correction pairs [6], trajectory fusion [10], and hint-guided recovery [1], were developed for mathematics and software engineering, and offer little guidance when an error depends on medical images, anatomy, or differential diagnosis. Moreover, a model's failures reveal which medical capabilities need further training, yet this signal is rarely used to guide the next round of supervision. We therefore ask two questions: how can we repair reasoning while **preserving valid steps**, and how can **failures guide subsequent learning**?

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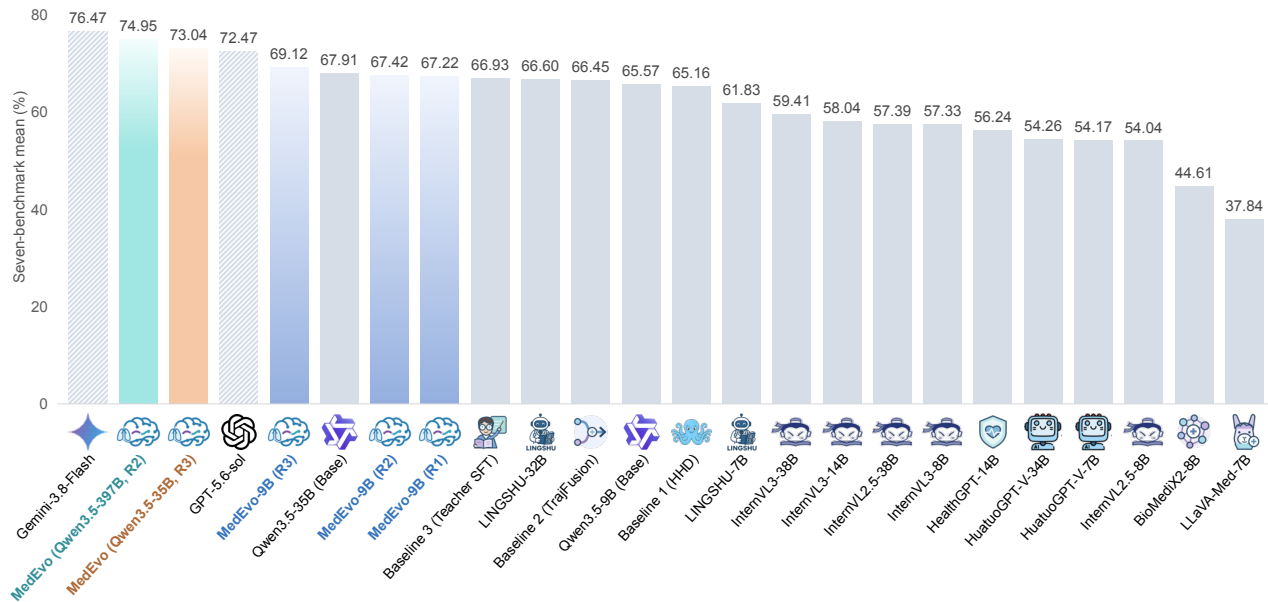


Figure 1 Average accuracy across seven medical multimodal benchmarks. The MedEvo models with 9B, 35B, and 397B parameters achieve strong performance across model scales, with MedEvo-9B outperforming several larger general-purpose and medical models in this comparison. Abbreviations: billion parameters (B), supervised fine-tuning (SFT), and Hindsight Hint Distillation (HHD) [1].

We introduce MedEvo through the principle of **Knowledge Navigation**: using structured medical knowledge and failure diagnosis to guide **where to intervene, how to recover, and what to learn next**. This definition follows from the two questions above: improving reasoning requires both a local recovery decision and a learning decision that reflects the student’s remaining weaknesses. Medical tags locate a failure in its clinical context, while error types identify the reasoning operation that needs correction. We call this navigation because these signals direct a path from an observed failure to a targeted intervention, at two connected levels: **within a trajectory** and **across training rounds** (Fig. 4). For a failed rollout, a gold-blind auditor identifies the first consequential error, preserves the valid prefix, and provides a **minimal, answer-free hint**; the same student then regenerates the erroneous suffix, so the correction remains grounded in its own reasoning (Fig. 2). After each round, MedEvo measures performance and errors across medical tags on a **held-out feedback set** and prioritizes questions for fresh rollouts and repairs, so that recurring failures shape subsequent training.

Across seven medical multimodal benchmarks (Fig. 1), MedEvo improves the mean accuracy of Qwen3.5-9B, Qwen3.5-35B, and Qwen3.5-397B from 65.57%, 67.91%, and 71.83% to **69.12%, 73.04%, and 74.95%**, respectively. With Qwen3.5-9B, it outperforms teacher-distillation SFT (69.12% vs. 66.93%); with Qwen3.5-35B and Qwen3.5-397B, it surpasses GPT-5.6-sol (72.47%); Gemini-3.8-Flash has the highest mean accuracy in this comparison (76.47%).

Our contributions are threefold:

- **Medically informed local repair.** Within trajectories, Knowledge Navigation uses a medical tag taxonomy and error-type diagnosis to locate the first consequential error and a safe rollback boundary; a gold-blind auditor provides an answer-free hint, and the same student regenerates the erroneous suffix while preserving its valid prefix.
- **Feedback-driven data evolution.** Across rounds, Knowledge Navigation uses feedback: capability gaps across medical tags and error types on a held-out feedback set prioritize the questions and failures used for fresh rollouts and repairs.
- **Empirical gains and findings.** With SFT alone, MedEvo improves Qwen3.5-9B, Qwen3.5-35B, and Qwen3.5-397B across seven benchmarks. Our analyses show that failed trajectories contain reasoning worth preserving, that repair success does not imply learning value, and that independent feedback is key to sustained improvement.

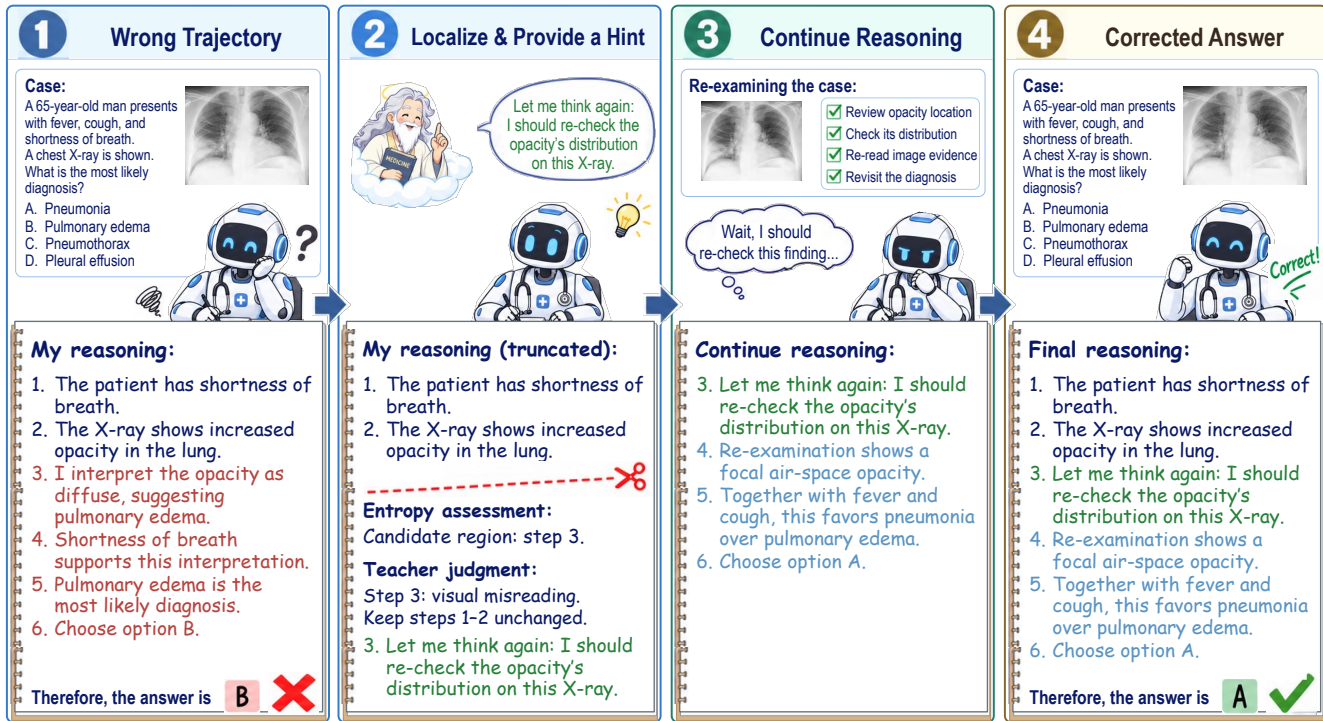


Figure 2 Local navigation from a failed reasoning trajectory. A student produces an incorrect answer; a teacher retains a supported prefix and supplies an answer-blind hint; the same student re-examines the evidence and continues to a corrected answer. The comic illustrates the interface; localization and admission checks are detailed in §3.2.1 and §3.2.4, respectively.

2 Related Work

2.1 Medical Multimodal Reasoning and Supervision Quality

Medical vision-language models have been built through biomedical instruction tuning [2, 3], generalist medical training in Lingshu [11], and bilingual image understanding in BiMediX2 [12]. MedGemma provides medical vision-language foundation models based on Gemma 3 [13]. Beyond answer accuracy, **process-level audits** separate visual recognition, knowledge application, and evidence integration [14, 15], and controlled hallucinations can escape rubric-based evaluation, motivating **explicit factuality checks** [16].

Supervision quality also depends on **reasoning content and student compatibility**: process supervision provides step-level feedback [17], on-policy distillation studies student-teacher compatibility [9], and ChainPrune [18] reduces redundant reasoning while distinguishing structural efficiency from token length. Building on these observations, MedEvo diagnoses failures with a medical error codebook and admits a repair only if it passes answer, boundary, integrity, and student-relative learnability checks.

2.2 Learning from Failures and Localized Repair

Failed trajectories can supply supervision: LEMA trains on mistake-correction pairs [6], TrajFusion combines incorrect and correct trajectories [10], and HHD guides recovery with concise hints [1]. Identifying an error can enable its correction [5], and backtracking distillation and selective regeneration study where correction should begin [19, 20]. MedEvo differs in three respects: the corrected suffix is **generated by the same student** rather than written by a teacher; the auditor **never accesses the reference answer**; and both localization and hints are conditioned on medical tags and error types.

2.3 Data Evolution and Recursive Self-Improvement

STaR and ReST^{EM} iteratively build supervision from self-generated reasoning filtered by outcome correctness [7, 8], and micro-reflective self-distillation adds capability-dependent trajectory selection [21]. MediSkill-Evo [22] instead evolves validated clinical knowledge banks without updating the backbone. MedEvo follows the self-training loop of STaR and ReST^{EM} but differs in two respects: rather than STaR’s rationalization, which supplies the correct answer as a hint [7], it repairs failures with answer-free hints; and it selects each round’s data using **medical-tag**

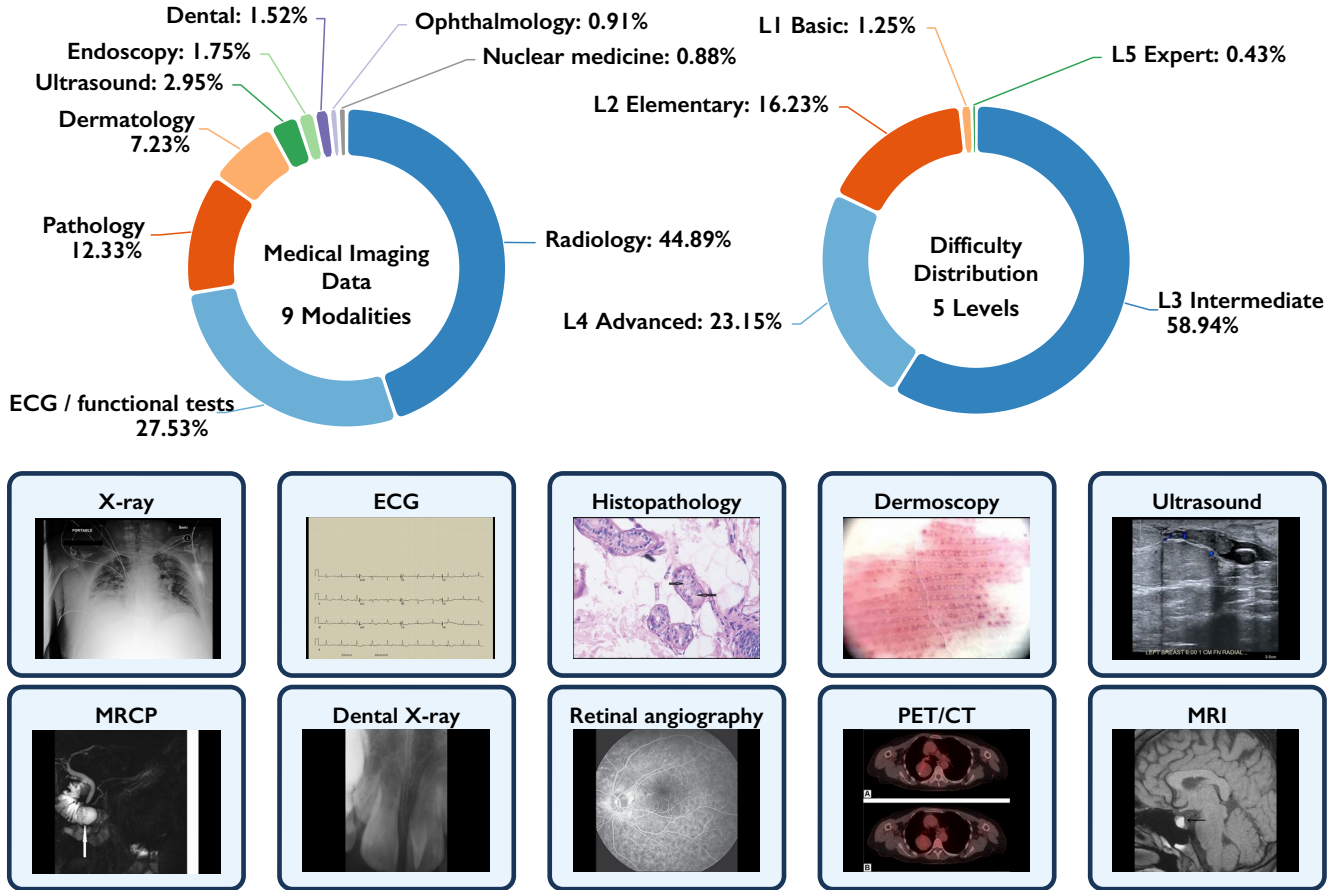


Figure 3 Overview of our medical multimodal data. Top: distributions across nine modality categories (left) and five difficulty levels (right). Bottom: representative images and functional-test examples illustrating the range of visual inputs. Abbreviations: electrocardiogram (ECG), magnetic resonance cholangiopancreatography (MRCP), positron emission tomography / computed tomography (PET/CT), and magnetic resonance imaging (MRI).

and error-type feedback on a fixed feedback set, linking local recovery to what the student should learn next. Knowledge Navigation couples medically informed local repair with feedback-guided data selection.

See Appendix F for additional related work on medical hallucinations, reasoning supervision, correction methods, and broader self-evolution settings.

3 Method

Knowledge Navigation. We define **Knowledge Navigation** as a feedback-guided process that uses structured domain knowledge and failure diagnosis to choose **where to intervene**, **how to recover**, and **what to learn next**. MedEvo instantiates this process with a medical tag taxonomy and an error codebook: tags organize the clinical context, error types identify the failed reasoning operation, and their associated recovery operations guide intervention. Within a trajectory, the process selects a safe rollback boundary and an answer-free hint while preserving supported reasoning (§3.2). Across rounds, measured capability gaps guide question selection and repair priorities (§3.3). The term navigation therefore describes the link from a structured diagnosis to an actionable decision and its subsequent reassessment. The taxonomy supplies the reference structure; the student’s changing failures determine the direction of learning (Fig. 4).

Learning protocol. A question $x_i = (I_i, q_i, O_i)$ consists of one or more medical images I_i , the clinical context and question q_i , and the answer options O_i . Following STaR and ReST^{EM} [7, 8], each round t uses the current student $M_t = \pi_{\theta_t}$ to generate trajectories, collects correct rollouts and accepted repairs into the training set D_{t+1} , and fine-tunes the next student from the original model M_0 rather than from M_t . The training pool, feedback set, and test benchmarks are kept separate throughout.

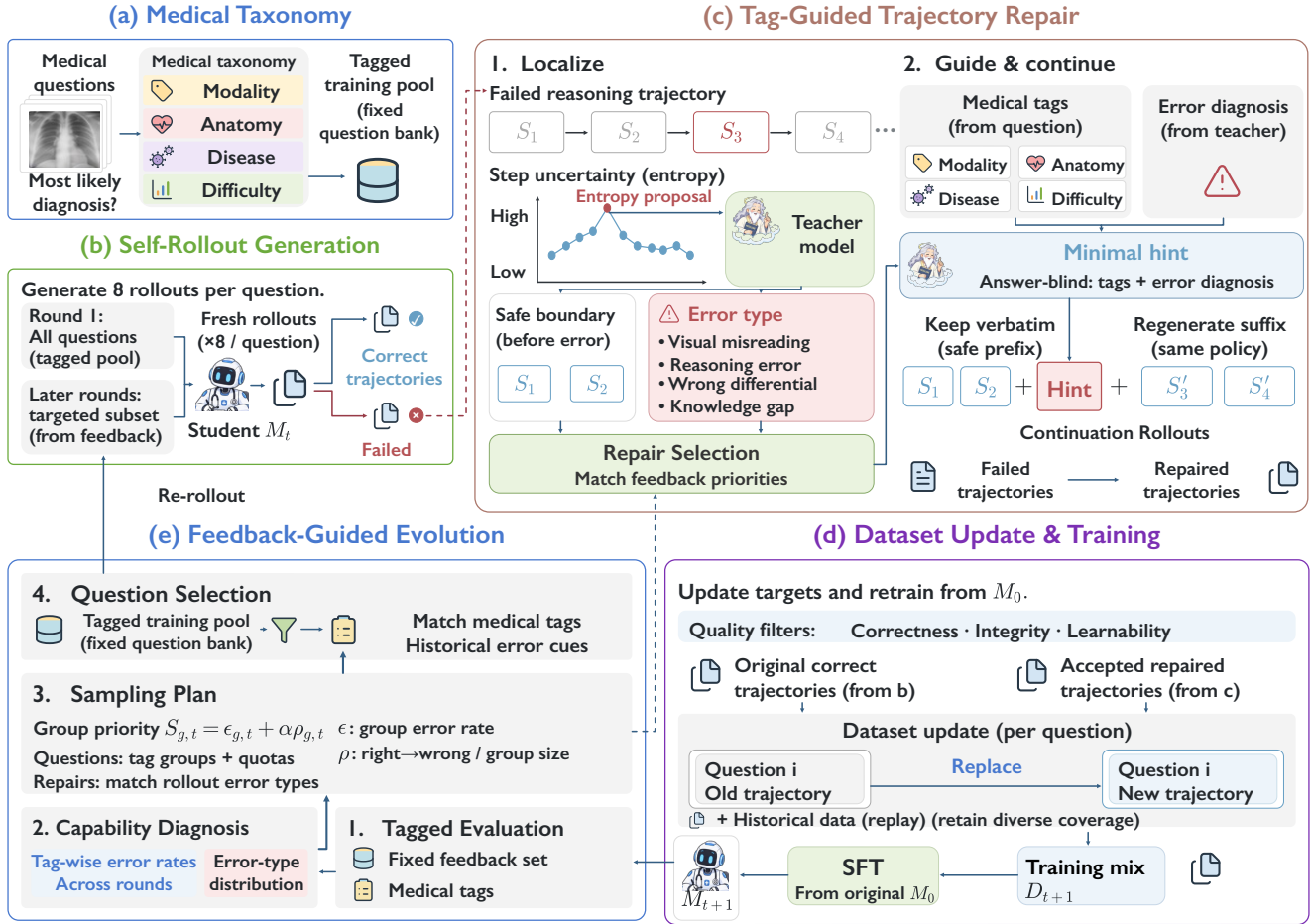


Figure 4 MedEvo overview of Knowledge Navigation within trajectories and across training rounds. (a) Medical questions are tagged once by modality, anatomy, disease, and difficulty. (b) The current model generates fresh rollouts for all questions in the first round and feedback-selected questions in later rounds. (c) A gold-blind auditor localizes failures and provides tag-guided minimal hints; the same student preserves a safe prefix and regenerates the suffix. (d) Correct and accepted repaired trajectories update the dataset by question, retaining valid historical targets and replacing them only with accepted improvements, before training the next model from M_0 . (e) Fixed-set evaluation, capability diagnosis, data planning, and targeted question retrieval guide the next rollout round.

3.1 Medical Taxonomy and Failure Diagnosis

3.1.1 Medical Question Tags

We assign each question a four-dimensional hierarchical tag

$$z_i = (z_i^{\text{mod}}, z_i^{\text{anat}}, z_i^{\text{dis}}, z_i^{\text{diff}}), \quad (1)$$

covering modality, anatomy, disease, and difficulty. Tags serve two roles. During repair, the auditor receives only the **observable subset** z_i^{obs} , i.e., tags supported by the input itself, excluding answer-derived labels such as the diagnosed disease. Across rounds, tag-level accuracy on the feedback set identifies capability gaps and guides question selection. See Appendix C.1 for the tag hierarchy and annotation rules.

3.1.2 Error Types and Their Relation to Repair

We diagnose failures at the process level, building on work that separates medical reasoning from knowledge and audits intermediate steps [14, 15]. Our codebook defines **four parent error types** with five subtypes each, twenty in total (Appendix C.1.2). Each type is paired with a recovery operation used for hint construction (§3.2.2): visual misreading calls for perceptual re-examination; reasoning error for reconstructing the evidence-to-decision link; differential diagnosis error for comparing discriminative features; and knowledge gap for a narrowly scoped knowledge cue. Whereas tags describe the question, error types describe what went wrong in a specific trajectory; because diagnoses vary across rollouts and checkpoints, **each fresh failure is diagnosed independently**.

3.2 Localized Repair of Failed Student Trajectories

3.2.1 Student Rollouts and Uncertainty-Based Localization

Tyen et al. [5] show that LLMs can correct reasoning errors once the error location is given; we therefore **localize an error before regenerating**. For each selected question, M_t samples eight trajectories, and answer-correct ones proceed to quality screening (§ 3.2.4). An incorrect or unparsable trajectory is segmented into semantically complete steps

$$r^- = (s_1, \dots, s_K), \quad (2)$$

such as image observation, feature interpretation, mechanism, differential diagnosis, and option mapping. The goal is to keep the steps before the **first consequential error** and regenerate the rest.

To propose where to audit, we compute the student’s token entropy

$$H_j = - \sum_{v \in \mathcal{V}} \pi_{\theta_t}(v \mid x, r_{<j}^-) \log \pi_{\theta_t}(v \mid x, r_{<j}^-), \quad (3)$$

where $\mathcal{V}$ is the vocabulary, j indexes tokens, and $r_{<j}^-$ is the token prefix. Logs are natural, with $0 \log 0 = 0$. For each nonempty step s_k , let $\mathcal{J}_k$ be its token-index set and $H_{s_k} = (H_j)_{j \in \mathcal{J}_k}$. We combine their **mean and 90th percentile** $Q_{0.9}$, and measure the increase over the preceding $m_k = \min(m, k - 1)$ steps:

$$E_k^* = \lambda \frac{1}{|\mathcal{J}_k|} \sum_{j \in \mathcal{J}_k} H_j + (1 - \lambda) Q_{0.9}(H_{s_k}), \quad (4)$$

$$\Delta E_k = \begin{cases} 0, & k = 1, \\ E_k^* - \frac{1}{m_k} \sum_{\ell=k-m_k}^{k-1} E_\ell^*, & 2 \leq k \leq K, \end{cases} \quad (5)$$

For $k \geq 2$, $m_k > 0$. After min-max normalizing both profiles (constant profiles map to zero), we select the anchor

$$k^* \in \arg \max_{1 \leq k \leq K} [\widetilde{E}_k^* + \widetilde{\Delta E}_k], \quad (6)$$

and the audit window covers steps $k^* \pm 1$, clipped to valid indices. We use $\lambda = 0.5$ and $m = 3$ (Appendix A.2). **Entropy proposes the audit region**; the auditor determines the error location and repair boundary from the images, question, and reasoning context (§ 3.2.2).

3.2.2 Gold-Blind Auditing and Medical Error Diagnosis

Building on hint-guided recovery in HHD [1], an external auditor $\mathcal{A}$ (GPT-5.6-sol) turns the diagnosis into concise guidance. The auditor is gold-blind: it receives the question and images, the failed trajectory, the entropy proposal $\mathcal{E}_t(x, r^-)$, and the observable tags z^{obs} , but **no reference answer, reference rationale, or answer-derived metadata**. It returns

$$(a, e, b, h) = \mathcal{A}(x, r^-, \mathcal{E}_t(x, r^-), z^{\text{obs}}), \quad (7)$$

where $a \in \{\text{REPAIR}, \text{SKIP}\}$ is the action. For REPAIR, e is the **earliest consequential error** together with its diagnosed type, b is the last fully valid step before e , and h is a **minimal recovery hint**. For SKIP, (e, b, h) are unused and no continuation is generated. The hint instantiates the recovery operation of the diagnosed type (§ 3.1.2) in the observable context. For example, a perceptual hint may ask the student to reinspect a region or waveform attribute without stating the finding, and a differential hint may request comparison of a discriminative feature without naming the diagnosis. Hints must not reveal the answer, an answer option, or a corrected rationale. The audit prompt appears in Appendix D.3.

3.2.3 Prefix Preservation and Same-Student Continuation

Like LEMA [6], we train on corrected mistakes, but the correction is generated by the student itself. The boundary splits the failed trajectory into a **preserved prefix and an erroneous suffix**,

$$r^- = p_b \oplus s^-, \quad p_b = (s_1, \dots, s_b). \quad (8)$$

Here $b \in \{0, \dots, K - 1\}$, p_0 is empty, and $\oplus$ denotes token-sequence concatenation. **We keep p_b verbatim**, append the hint rendered as a natural-language recovery cue $g(h)$, and let the same checkpoint continue:

$$s^+ \sim \pi_{\theta_t}(\cdot \mid x, p_b \oplus g(h)), \quad r^* = p_b \oplus g(h) \oplus s^+. \quad (9)$$

The question stays in the user message, $p_b \oplus g(h)$ prefills an unfinished assistant message, and the structured audit is stored only as metadata. With the prefix fixed, we sample up to five suffixes per failed trajectory, **stopping at the first admitted continuation**, and send answer-correct continuations to admission (§3.2.4; Appendix D.4).

3.2.4 Repair Admission and Training Targets

A continuation is admitted only if it passes **answer, boundary, integrity, and learnability checks**. Learnability is scored on the **new suffix alone**, so that a valid prefix cannot mask the difficulty of the correction. Adapting the instruction-following difficulty of Li et al. [23], we measure how much the suffix depends on the task input:

$$D_{\text{repair}} = \frac{\mathcal{L}_t(s^+ | x, p_b)}{\mathcal{L}_t(s^+ | p_b)}, \quad (10)$$

For a token sequence s and context c , $\mathcal{L}_t(s | c) = -\sum_{j=1}^{|s|} \log \pi_{\theta_t}(s_j | c, s_{<j})$, in nats. Both terms are **computed without the hint**; the denominator omits x but retains p_b . A smaller ratio indicates a larger relative loss reduction from explicitly conditioning on x beyond the prefix, which may itself contain task information. Both suffixes must be nonempty, all losses finite, and the denominator strictly positive. We also measure how strongly the student still prefers its original wrong suffix, using per-token losses to account for length differences:

$$\Delta_{\text{repair}} = \frac{\mathcal{L}_t(s^+ | x, p_b)}{|s^+|} - \frac{\mathcal{L}_t(s^- | x, p_b)}{|s^-|}. \quad (11)$$

For a repair candidate $\xi = (x, r^-, r^*, b)$, the gap is measured in nats per token. A positive value means the student assigns lower average loss to the wrong suffix than to the repair. The learnability gate L_ξ requires $D_{\text{repair}} \in [0.35, 0.85]$ and $\Delta_{\text{repair}} \in [0.02, 0.30]$, with **thresholds calibrated on the feedback set** and frozen before final evaluation (Appendix A.4). Admission is

$$\text{Accept}(\xi) = C_\xi \wedge A_\xi \wedge N_\xi \wedge L_\xi, \quad (12)$$

where C_ξ verifies the answer against the reference label, A_ξ checks the **preserved prefix and boundary**, and N_ξ rejects unsupported additions and invalid repetition.

Each admitted repair is trained as the complete response $r^* = p_b \oplus g(h) \oplus s^+$, including the recovery cue, with **loss on all assistant tokens**:

$$\mathcal{L}_{\text{repair}}(\theta) = -\sum_{j=1}^{|r^*|} \log \pi_{\theta}(r_j^* | x, r_{<j}^*). \quad (13)$$

Original correct rollouts use the same objective, and image and user tokens are excluded. If no repair is admitted, we keep a qualified correct rollout or the existing valid target; otherwise no target is added. **No hints or auditor outputs are used at evaluation.**

3.3 Feedback-Driven Evolution of Supervision

Across-round Knowledge Navigation follows the iterative self-training pattern of STaR and ReST^{EM} [7, 8], but uses **medical-tag and error-type feedback on a fixed set** to allocate generation and repair effort. The cycle is assessment, targeted selection, local repair, screened dataset update, and retraining. The taxonomy and error codebook remain fixed; **selection priorities and the supervision dataset evolve** as the student changes.

3.3.1 First-Round Coverage

The training pool $\mathcal{T}$ and a separate, fixed feedback set V_{feedback} are drawn from public medical datasets (§4.1). In the first round, M_0 generates **rollouts for every question** in $\mathcal{T}$; correct trajectories enter screening, failures are repaired as in §3.2, and accepted targets form D_1 , giving $M_1 = \text{SFT}(M_0, D_1)$.

3.3.2 Feedback across Medical Tags and Error Types

Before constructing D_{t+1} , we evaluate M_t on the feedback set, including M_0 at initialization. For a supported tag group g with $n_g > 0$, the error rate is

$$\epsilon_{g,t} = \frac{1}{n_g} \sum_{i \in V_{\text{feedback},g}} (1 - c_{i,t}), \quad n_g = |V_{\text{feedback},g}|, \quad (14)$$

where $c_{i,t} \in \{0, 1\}$ indicates correctness. Comparing consecutive checkpoints separates corrections, regressions, persistent errors, and retained successes (Appendix A.5), and diagnosing failed responses yields each group's

error-type distribution. Groups below a support threshold back off to their parent tag. Feedback questions guide selection and threshold calibration but **never enter training**, and test results never influence data construction.

3.3.3 Selecting Questions and Updating the Dataset

From the second round on, we allocate the generation budget across groups by the rule-based priority

$$S_{g,t} = \epsilon_{g,t} + \alpha \rho_{g,t}, \quad \rho_{g,t} = \frac{N_{g,t}^{10}}{n_g}, \quad (15)$$

where $t \geq 1$ and $N_{g,t}^{10}$ counts correct-to-incorrect transitions from M_{t-1} to M_t and $\alpha \geq 0$ weights regressions. Quotas are proportional to $S_{g,t}$, capped by the available unique questions, and rounded deterministically; unused capacity is redistributed, and each multi-tag question is counted once (Appendix A.5). Within each group, candidates are retrieved by tag match and ranked by historical error records. Selected questions receive **fresh rollouts** from M_t , their failures are diagnosed anew, and failures whose error types match the feedback priorities are repaired first. Feedback thus decides where effort is spent, while the checks in §3.2.4 decide what enters training.

The dataset keeps **at most one active target per question**. An admitted candidate replaces the existing target only if a blinded comparison judges it better in evidence support, reasoning consistency, and conciseness; otherwise the existing target is kept, and all versions are logged separately. Writing each target as a pair (x, r) keyed by question ID, with $D_0 = \emptyset$, replaced targets $R_t \subseteq D_t$, accepted replacements U_{t+1} , and new targets A_{t+1} for previously unrepresented questions,

$$D_{t+1} = (D_t \setminus R_t) \cup U_{t+1} \cup A_{t+1}, \quad (16)$$

$$M_{t+1} = \text{SFT}(M_0, D_{t+1}). \quad (17)$$

U_{t+1} contains exactly one replacement for each question in R_t ; A_{t+1} contains one target per new question, disjoint by question ID from D_t . Each round restarts from M_0 with a fresh optimizer and learning-rate schedule, so **experience is carried only through the evolving dataset**. Algorithm 1 summarizes the procedure.

4 Experiments

4.1 Experimental Setup

Models and data. Our main backbone is Qwen3.5-9B [24]; Qwen3.5-35B and Qwen3.5-397B test scalability (§5.3). The training pool contains 81,293 questions from the training splits of public medical datasets, spanning nine modality categories and five difficulty levels (Fig. 3), and is deduplicated by question and source IDs, patient/case IDs when available, image hashes, and near-duplicate clusters (Appendix C.1.4). A separate, fixed set of 5,000 modality-stratified questions serves as the feedback set and **never enters training**. Dataset roles and training configurations are given in Appendices B.1 and C.1.5.

Table 1 Round-wise repair statistics. Success rate = successfully repaired / sent for repair. Arrows indicate changes from the preceding round.

Training round	R1	R2	R3
Generator	R0	R1	R2
Sent for repair	27,233	25,831↓	23,620↓
Share of question bank	33.50%	31.78%	29.06%
Successfully repaired	19,567	18,258	16,797
Repair success rate	71.85%	70.68%↓	71.11%↑
Mean repair attempts	3.37	2.48	2.12

Evaluation. We use MedUniEval [25], the Hulu-Med implementation built on Lingshu’s MedEvalKit [11], on PMC-VQA [26], VQA-RAD [27], SLAKE [28], PathVQA [29], OmniMedVQA [30], the multimodal track of MedXpertQA (MedXpertQA-MM) [4], and the medical subset of MMMU (MMMU-Med) [31]. All models share the **answering prompt and parser** (Appendix D.1). We report accuracy and the unweighted seven-benchmark mean.

Baselines. On Qwen3.5-9B, we compare with two methods that learn from failed trajectories, HHD [1] and TrajFusion [10], and with **teacher-distillation SFT** on rationales generated by Qwen-3.8-Max. External general-purpose and medical models provide additional context (Appendix G).

4.2 Main Results

Table 2 reports the results. On Qwen3.5-9B, MedEvo raises mean accuracy from 65.57% to 69.12% and **improves all seven benchmarks**, with the largest gain on MedXpertQA-MM [4] (47.40% → 54.50%). It outperforms HHD [1], TrajFusion [10], and teacher-distillation SFT, whose mean accuracies are 65.16%, 66.45%, and 66.93%, respectively. On

Algorithm 1: MedEvo Training with Local Repair and Feedback-Guided Data Evolution

Require: Initial student M_0 ; training pool $\mathcal{T}$ with verification labels; fixed feedback set V_{feedback} ; gold-blind auditor $\mathcal{A}$; rounds R ; selection and generation budgets.

$\mathcal{G} \leftarrow \text{AssignMedicalTags}(\mathcal{T}, V_{\text{feedback}})$

$D_0 \leftarrow \emptyset; H \leftarrow \emptyset$ *// Active targets and trajectory history*

for $t \leftarrow 0$ **to** $R - 1$ **do**

// Feedback assessment and question selection

$F_t \leftarrow \text{Evaluate}(M_t, V_{\text{feedback}}, \mathcal{G})$

if $t = 0$ **then** *// Broad first-round coverage*

$Q_t \leftarrow \mathcal{T}$

else

$P_t \leftarrow \text{PlanQuotas}(F_t, F_{t-1}, \mathcal{G})$

$Q_t \leftarrow \text{RetrieveUnique}(\mathcal{T}, P_t, H)$

$D_{t+1} \leftarrow D_t$

// Fresh student rollouts and local trajectory repair

foreach $x \in Q_t$ **do**

$\mathcal{R}_x \leftarrow \text{Rollout}(M_t, x, 8)$

$C_x \leftarrow \text{ScreenCorrect}(x, \mathcal{R}_x)$

foreach failed $r^- \in \text{PrioritizeFailures}(\mathcal{R}_x, F_t)$ within budget **do**

$u \leftarrow \text{StepUncertainty}(M_t, x, r^-)$

$(a, e, b, h) \leftarrow \mathcal{A}(x, r^-, u, z^{\text{obs}}(x))$ *// Observable tags only*

$H \leftarrow \text{RecordHistory}(H, x, r^-, a, e, b, h)$

if $a = \text{SKIP}$ **then**

continue

$p_b \leftarrow \text{ValidPrefix}(r^-, b)$

for $j \leftarrow 1$ **to** 5 **do**

$s^+ \sim M_t(\cdot \mid x, p_b \oplus g(h))$

$r^* \leftarrow p_b \oplus g(h) \oplus s^+$

$\xi \leftarrow (x, r^-, r^*, b)$

if $\text{Accept}(\xi)$ (Eq. 12) **then**

$C_x \leftarrow C_x \cup \{r^*\}$

break

// Keep one active target per question; retain valid history on ties

$D_{t+1} \leftarrow \text{RetainOrReplace}(D_{t+1}, x, C_x)$

// Full-response SFT from the same initial student

$M_{t+1} \leftarrow \text{SFT}(M_0, D_{t+1})$ *// Reset optimizer and schedule*

return $\{M_t, D_t\}_{t=1}^R$

Qwen3.5-35B, mean accuracy rises from 67.91% to 73.04% (§5.3). A question-level paired bootstrap (10,000 resamples; Table 3, right) yields **95% CIs that exclude zero** on every benchmark, including [2.4, 4.7] percentage points (pp) for the mean gain and [5.2, 9.0] for MedXpertQA-MM. These intervals capture question-level uncertainty; variation across training runs is reported in Table 8.

4.3 Round-wise Evolution

Mean accuracy **increases monotonically across rounds**, from 65.57% to 67.22%, 67.42%, and 69.12% (Table 3, left); §5.2 isolates the contribution of feedback-guided rounds to this improvement. Individual benchmarks fluctuate, e.g., PathVQA [29] (52.38%, 51.27%, 53.34%). Meanwhile, the repair workload falls from 27,233 to 23,620 questions while the success rate stays near 71% (Table 1; Appendix B.2). On the fixed feedback set, every modality group improves from R0 to R3, with accuracy gains ranging from 4.50 to 14.07 pp for functional tests and ophthalmology, respectively; per-round modality and error-type statistics are given in Appendix B.3. Feedback-set trends guide subsequent data selection; the **seven held-out benchmarks measure generalization**.

Table 2 Accuracy (%) on seven medical benchmarks.  denotes medical-specialized models;  denotes trajectory-repair methods (HHD, TrajFusion, and MedEvo). MedEvo combines medical specialization with trajectory repair. Both MedEvo models are evaluated after Round 3. Avg. is the unweighted seven-benchmark mean; MedXQA is the table abbreviation for MedXpertQA-MM; MMMU-Med denotes the medical subset of MMMU. Blue shading highlights MedEvo. Bold and underlining mark the best and second-best values in each column within each model group; ties share the same rank.

Model	PMC-VQA	VQA-RAD	SLAKE	PathVQA	OmniMed VQA	MedXQA	MMMU-Med	Avg.
<i>Proprietary models</i>								
GPT-5.6-sol	<u>60.43</u>	<u>70.29</u>	81.34	<u>54.84</u>	<u>76.67</u>	<u>77.50</u>	<u>86.24</u>	<u>72.47</u>
Gemini-3.8-Flash	64.52	74.72	81.34	58.58	80.76	86.30	89.10	76.47
<i>Open-source models (< 10B)</i>								
 LLaVA-Med-7B [2]	30.50	53.70	48.00	38.80	44.30	20.30	29.30	37.84
 HuatuoGPT-V-7B [32]	53.30	67.00	67.80	48.00	74.20	21.60	47.30	54.17
 BiMediX2-8B [12]	43.50	49.20	57.70	37.00	63.30	21.80	39.80	44.61
InternVL2.5-8B [33]	51.30	59.40	69.00	42.10	81.30	21.70	53.50	54.04
InternVL3-8B [34]	53.80	65.40	72.80	48.60	79.10	22.40	59.20	57.33
 LINGSHU-7B [11]	56.30	67.90	83.10	61.90	82.90	26.70	54.00	61.83
Qwen3.5-9B (Base) [24]	59.20	68.10	76.80	48.80	82.20	47.40	76.50	65.57
 Baseline 1 (HHD) [1]	58.93	67.65	71.45	50.12	82.35	48.75	76.90	65.16
 Baseline 2 (TrajFusion) [10]	59.32	69.12	77.23	49.23	<u>83.46</u>	49.56	77.20	66.45
Baseline 3 (Teacher SFT)	<u>61.23</u>	<u>70.15</u>	71.60	53.20	<u>81.90</u>	<u>52.30</u>	<u>78.14</u>	<u>66.93</u>
  MedEvo (9B)	62.68	70.40	<u>78.23</u>	<u>53.34</u>	85.41	54.50	79.28	69.12
<i>Open-source models (> 10B)</i>								
 HealthGPT-14B [35]	56.40	65.00	66.10	<u>56.70</u>	75.20	24.70	49.60	56.24
 HuatuoGPT-V-34B [32]	56.60	61.40	69.50	44.40	74.00	22.10	51.80	54.26
InternVL3-14B [34]	54.10	66.30	72.80	48.00	78.90	23.10	63.10	58.04
InternVL2.5-38B [33]	57.20	61.40	70.30	46.90	79.90	24.40	61.60	57.39
InternVL3-38B [34]	56.60	65.40	72.70	51.00	79.80	25.20	65.20	59.41
 LINGSHU-32B [11]	57.90	76.50	89.20	65.90	<u>83.40</u>	30.90	62.30	66.59
Qwen3.5-35B (Base) [24]	<u>62.40</u>	70.50	70.10	49.15	<u>83.20</u>	<u>60.00</u>	<u>80.00</u>	<u>67.91</u>
  MedEvo (35B)	64.63	<u>75.83</u>	<u>80.30</u>	56.01	83.97	66.50	84.02	73.04

Table 3 Round-wise accuracy and paired bootstrap gains for the model based on Qwen3.5-9B. Left: accuracy (%) across R0–R3; each round retrains from Base (R0). Bold and underlining mark the best and second-best values per benchmark; arrows compare consecutive rounds, with MedXQA changes in pp. Right: R3 minus R0 and its 95% confidence interval (CI), in pp, from 10,000 question-level paired bootstrap resamples. Avg. is the unweighted seven-benchmark mean; its CI is computed by resampling independently within each benchmark before averaging. All gain intervals exclude zero.

Benchmark	Round-wise accuracy				Paired bootstrap	
	R0	R1	R2	R3	Δ (R3–R0)	95% CI
PMC-VQA	59.20	59.54 $\uparrow$	<u>60.97</u> $\uparrow$	62.68 $\uparrow$	+3.48	[+1.9, +5.1]
VQA-RAD	68.10	<u>68.47</u> $\uparrow$	68.06 $\downarrow$	70.40 $\uparrow$	+2.30	[+0.8, +3.8]
SLAKE	76.80	<u>76.92</u> $\uparrow$	<u>77.34</u> $\uparrow$	78.23 $\uparrow$	+1.43	[+0.2, +2.7]
PathVQA	48.80	<u>52.38</u> $\uparrow$	51.27 $\downarrow$	53.34 $\uparrow$	+4.54	[+2.6, +6.5]
OmniMedVQA	<u>82.20</u>	80.81 $\downarrow$	81.58 $\uparrow$	85.41 $\uparrow$	+3.21	[+1.5, +4.9]
MedXQA	47.40	52.56 $\uparrow$ 5.16	<u>52.79</u> $\uparrow$ 0.23	54.50 $\uparrow$ 1.71	+7.10	[+5.2, +9.0]
MMMU-Med	76.50	<u>79.83</u> $\uparrow$	79.91 $\uparrow$	79.28 $\downarrow$	+2.78	[+1.3, +4.3]
Avg.	65.57	<u>67.22</u> $\uparrow$	<u>67.42</u> $\uparrow$	69.12 $\uparrow$	+3.55	[+2.4, +4.7]

5 Ablation Studies and Analysis

We analyze repair design, feedback across rounds, model scale, and the **learning value of repair supervision**. All analyses except those on scale use Qwen3.5-9B [24].

5.1 Repair Design in the First Round

Table 4 compares four first-round repair designs that vary how much of the trajectory is regenerated (full rewrite vs. suffix continuation) and how specific the hint is (none, generic, or type-guided). We evaluate repair success and downstream learning using repair success rate and the resulting model’s mean accuracy. **Type-guided suffix continuation performs best** on both metrics, reaching 71.80% repair success and 67.22% mean accuracy. Generic-hint continuation achieves 64.50% repair success and 64.20% mean accuracy, while hint-guided full rewriting achieves 57.80% and 63.20%, respectively. These comparisons indicate that both prefix preservation and error-type conditioning contribute.

Table 4 First-round repair designs (9B). Both metrics are percentages; Base mean accuracy is 65.57%. Red ↓ marks lower values than the full method; bold marks the best values.

Method	Repair success	Mean accuracy
Hint + full rewrite	57.80↓	63.20↓
No-hint rewrite	36.70↓	61.16↓
Generic hint + suffix continuation	64.50↓	64.20↓
Type-guided hint + suffix continuation (Ours)	71.80	67.22

5.2 Contribution of Feedback across Rounds

Table 5 isolates the contribution of feedback. All variants start from the same Round-1 checkpoint (67.22%) and run two further rounds: repair without feedback, feedback without medical tags, feedback without error types, and full MedEvo. Full MedEvo reaches 69.12% at R3, compared with 66.67%, 67.53%, and 66.21% for the three variants, respectively. Without feedback or without error-type feedback, accuracy falls below the shared starting checkpoint; without medical tags, it improves only marginally (67.22% to 67.53%). **Combining medical-tag and error-type feedback yields the largest gain** in this comparison.

Table 5 Feedback ablation (9B). All runs share R1 = 67.22%. Arrows compare mean accuracy with R1; Δ is R3 minus R1 in pp.

Method	R2	R3	Δ
No feedback (repair only)	66.39↓	66.67↓	−0.55
Without medical tags	67.32↑	67.53↑	+0.31
Without error-type feedback	67.01↓	66.21↓	−1.01
Full MedEvo	67.42↑	69.12↑	+1.90

5.3 Generalization across Model Scales

We apply MedEvo to Qwen3.5-35B for three rounds and to Qwen3.5-397B for two rounds, evaluating on the same seven benchmarks [4, 26, 27, 28, 29, 30, 31] (Table 6). On Qwen3.5-35B, mean accuracy rises from 67.91% to 69.73%, 70.81%, and 73.04%. MedXpertQA-MM [4] peaks at 66.73% in R2 and declines slightly to 66.50% in R3, indicating benchmark-dependent optima. On Qwen3.5-397B, mean accuracy rises from 71.83% to 74.71% and 74.95%, and MedXpertQA-MM from 67.65% to 75.00%. The R2 model improves six of seven benchmarks, with OmniMedVQA [30] essentially unchanged (85.10% → 85.05%). At both scales, **mean accuracy increases in every round**.

5.4 Learning Value of Repair Supervision

Beyond repair success, we examine three questions about **what makes repair supervision useful** (Fig. 5).

Which trajectories are worth repairing? We compare the 8,000 questions with the longest failed trajectories against 8,000 random questions, each used for one round of continuation training. The longest set has lower repair success (55.70% vs. 67.50%) but yields higher downstream accuracy (66.78% vs. 64.35%; Fig. 5a), so trajectories should be selected for **downstream learning value**.

How many hints help? All main experiments provide one hint per failed trajectory. Using the longest-8k subset from the previous comparison, we vary the number of hints per failed trajectory from 1 to 3, 5, and 8, keeping five continuation attempts. Mean accuracy rises from 65.78% with one hint to 67.01% with three, then falls to 66.74% and 64.10% with five and eight (Fig. 5b), indicating that **moderate guidance helps** while excessive hints degrade the resulting supervision.

Which feedback guides evolution? Prioritizing repairs by training-set errors raises training accuracy from 63.23% to 78.00% but lowers benchmark accuracy from 65.57% to 64.01%, whereas the fixed feedback set yields 71.00%

Table 6 Round-wise accuracy (%) for models based on Qwen3.5-35B (left) and Qwen3.5-397B (right). R0 is Base; each later round retrains from the same Base. Bold and underlining mark the best and second-best results within each model scale per benchmark. Arrows show changes from the preceding round; numbers beside the highlighted MedXQA results give changes in pp. Avg. is the seven-benchmark mean.

Benchmark	35B				397B		
	R0	R1	R2	R3	R0	R1	R2
PMC-VQA	62.40	<u>63.12</u> ↑	60.33 ↓	64.63 ↑	63.47	<u>64.87</u> ↑	65.72 ↑
VQA-RAD	70.50	62.08 ↓	<u>70.91</u> ↑	75.83 ↑	74.72	<u>76.05</u> ↑	77.15 ↑
SLAKE	70.10	78.14 ↑	<u>79.18</u> ↑	80.30 ↑	77.38	<u>78.89</u> ↑	79.10 ↑
PathVQA	49.15	<u>52.71</u> ↑	50.87 ↓	56.01 ↑	52.49	58.51 ↑	<u>57.90</u> ↓
OmniMedVQA	83.20	85.39 ↑	<u>85.22</u> ↓	83.97 ↓	85.10	85.15 ↑	85.05 ↓
MedXQA	60.00	65.68 ↑ +5.68	66.73 ↑ +1.05	<u>66.50</u> ↓ -0.23	67.65	<u>74.40</u> ↑ +6.75	75.00 ↑ +0.60
MMMU-Med	80.00	81.02 ↑	<u>82.41</u> ↑	84.02 ↑	82.03	85.10 ↑	<u>84.75</u> ↓
Avg.	67.91	69.73 ↑	<u>70.81</u> ↑	73.04 ↑	71.83	<u>74.71</u> ↑	74.95 ↑

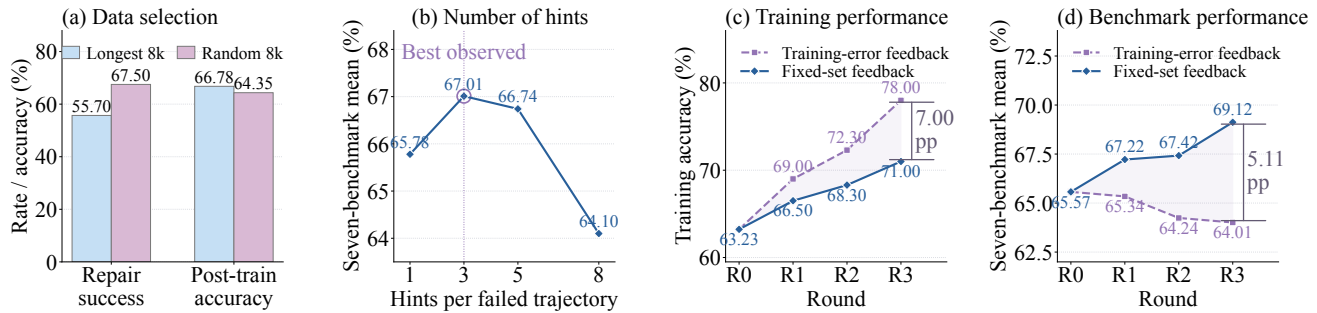


Figure 5 Trajectory selection, hint count, and feedback source for the 9B model. (a) Grouped bars compare longest-8k and random-8k selection in single-round training: longer trajectories have lower repair success but higher downstream accuracy. (b) Mean accuracy with 1, 3, 5, or 8 hints per failed trajectory on the longest-8k subset, keeping five continuation attempts; three hints perform best. (c,d) Training accuracy and independent seven-benchmark mean under training-error feedback versus a fixed 5,000-question feedback set. The feedback set guides selection but is excluded from training.

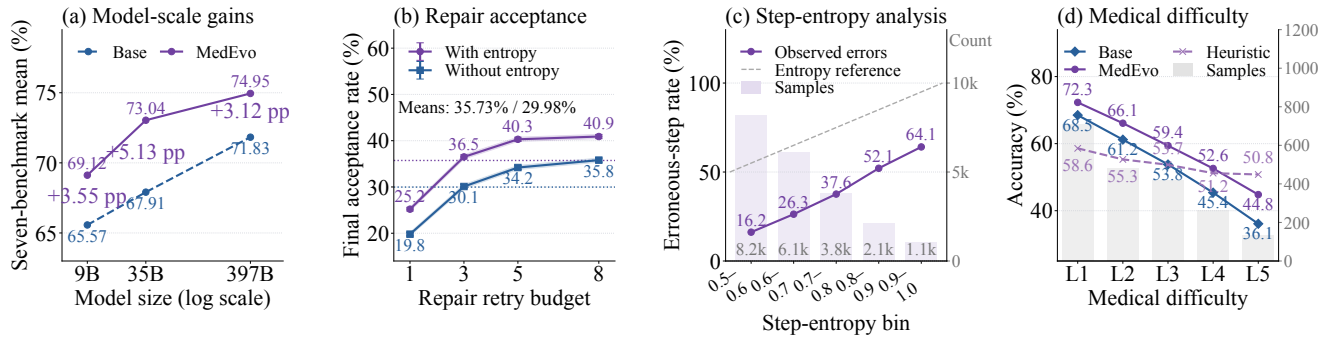


Figure 6 Analysis across scale, repair budget, entropy, and difficulty. (a) Seven-benchmark mean accuracies for the MedEvo models based on Qwen3.5-9B and Qwen3.5-35B after Round 3, and Qwen3.5-397B after Round 2. (b) Repair acceptance with and without entropy localization; intervals are pointwise 95% Wilson intervals using 45,000 attempts per setting. (c) Erroneous-step rates and step counts by entropy bin; the diagonal provides a numerical reference. (d) Accuracy by medical difficulty, with sample counts on the right axis.

training accuracy and 69.12% benchmark accuracy (Fig. 5c,d). Training failures supply repair candidates, but **an independent feedback set better identifies weaknesses that generalize**.

Further analyses. Fig. 6 summarizes additional repair analyses. Entropy localization increases final acceptance, defined as the fraction of repair candidates retained, at every continuation budget: from 19.8–35.8% to 25.2–40.9% for one to eight attempts (Fig. 6b). The erroneous-step rate rises from 16.2% to 64.1% across increasing entropy bins, supporting **entropy as an audit prior** (Fig. 6c). Gains over Base grow with difficulty, from 3.8 to 8.7 pp between L1 and L5 (Fig. 6d).

6 Discussion and Insights

We first examine the **reliability of repaired supervision** through a human audit and the **stability of MedEvo** across independent training runs. We then distill what these results, together with the analyses in §5, imply for medical self-evolution, and close with limitations and future directions.

6.1 Human Audit

Table 7 reports results for the audited sample of repaired trajectories. Annotators identify 95.5% of submitted trajectories as erroneous ($\kappa = 0.90$) and validate 84.5% of the selected repair boundaries ($\kappa = 0.85$). Hints are answer-blind in 97.2% of cases ($\kappa = 0.94$). Repaired answers are judged correct in 88.3% of cases ($\kappa = 0.89$), and 78.5% of repairs are judged **image-grounded**. Across the four

Table 7 Human audit pass rates (%) by error family: visual (V), reasoning (R), differential diagnosis (D), and knowledge (K). Agreement: κ with 95% CIs. Blue highlights repaired-output checks.

Audit criterion	Pass (%)	Agreement		Error family			
		κ	95% CI	V	R	D	K
Original trajectory erroneous	95.5	0.90	[0.83, 0.97]	96	96	94	96
Valid repair boundary	84.5	0.85	[0.66, 0.88]	88	84	82	84
No answer leakage in hint	97.2	0.94	[0.87, 1.00]	96	100	96	97
Image-grounded repaired reasoning	78.5	0.70	[0.58, 0.82]	76	80	76	82
Correct repaired answer	88.3	0.89	[0.81, 0.97]	88	89	88	88

error families, answer-correctness pass rates range from 88% to 89%, and grounding pass rates from 76% to 82%.

The audit assesses **answer correctness and visual grounding as complementary dimensions** of repair quality, reporting separate marginal pass rates of 88.3% and 78.5%, respectively. Agreement is higher for answer correctness ($\kappa = 0.89$) than for grounding ($\kappa = 0.70$). These results motivate evaluating both the final answer and the image evidence supporting intermediate deductions.

6.2 Stability across Random Seeds

Table 8 reports three independent training runs from the same Base checkpoint. The **improvement is highly reproducible**. Every run follows the ordering Base < R1 < R2 < R3, and R3 averages 69.09% $\pm$ 0.24, compared with 65.57% for Base. Even the weakest run reaches 68.84%, so the improvement far exceeds the variation across seeds. The round-wise pattern is equally consistent: R1→R2 yields a small accuracy gain (0.16–0.22 pp), followed by a larger gain at R2→R3 (1.63–1.71).

Each round retrains from M_0 , so **the evolving dataset carries information across rounds**. Across all three seeds, the R2-to-R3 gain exceeds the R1-to-R2 gain. Longer training runs can characterize the timing and saturation of these gains.

Table 8 Seven-benchmark mean accuracy (%) over three Qwen3.5-9B runs. SD is the sample standard deviation; Δ is gain over Base in pp. Bold/underlining mark best/second-best rounds; R3 is blue.

Model	Seed 1	Seed 2	Seed 3	Mean $\pm$ SD	Δ
Base	65.57	65.57	65.57	65.57 $\pm$ 0.00	—
R1	67.22	67.05	67.38	67.22 $\pm$ 0.17	$\uparrow$ 1.65
R2	<u>67.42</u>	<u>67.21</u>	<u>67.60</u>	<u>67.41 $\pm$ 0.20</u>	$\uparrow$ 1.84
R3	69.12	68.84	69.31	69.09 $\pm$ 0.24	$\uparrow$ 3.52

6.3 Insights for Medical Self-Evolution

Taken together, our results suggest that effective self-evolution in medical reasoning rests on three choices: **what reasoning to preserve, which repairs to admit as supervision, and which failures to revisit next**.

Defining characteristics of Knowledge Navigation. Three characteristics connect **medical context, failure diagnosis, and learning decisions**. First, navigation is **knowledge-structured**: tags and error types make a failure interpretable in terms of clinical context and a recoverable reasoning operation. Second, it is **selective and student-relative**: local repair preserves supported steps, and admission assesses whether a correction is suitable supervision for the current student. Third, it is **feedback-guided**: independent assessment redirects subsequent data selection as weaknesses change. Together, these properties connect the correction of one trajectory to the development of a broader capability. The repair and feedback ablations (Tables 4 and 5) support these design choices within MedEvo.

Evolution through supervision. Knowledge Navigation couples two scales of change. **Local recovery** turns an individual failure into a screened training target; **across-round adaptation** uses capability feedback to decide which targets should be generated next. The resulting cycle is assessment, targeted selection, local repair, dataset update, and reassessment after training. In the present implementation, the taxonomy and error codebook stay fixed, while selection priorities, accepted supervision, and the resulting student capabilities change. Each round retrains from M_0 on the updated dataset, so the evolving supervision carries information across rounds. This makes the evolution mechanism explicit and allows regressions and persistent weaknesses to redirect later effort.

Failed trajectories retain valuable reasoning. Suffix continuation outperforms full rewriting in both repair success and downstream accuracy (Table 4), supporting the value of **preserving the valid prefix**. In the trajectory-selection experiment, the longest-8k subset has lower repair success than random selection (55.70% vs. 67.50%) and yields higher downstream accuracy (66.78% vs. 64.35%; Fig. 5a). These comparisons support selecting repair candidates by their contribution to subsequent learning.

Repairability and learning value are different objectives. Repair success measures whether a correction is reached, whereas downstream accuracy measures what the student learns from it. Our experiments separate the two: harder-to-repair trajectories teach more (Fig. 5a), and guidance beyond a moderate number of hints lowers downstream accuracy (Fig. 5b). Repair should therefore provide minimal guidance and be **evaluated by its effect on subsequent learning**. The same division of labor applies to localization. Entropy reliably indicates where to inspect, with the erroneous-step rate rising from 16.2% to 64.1% across entropy bins (Fig. 6c), while the evidence-grounded audit determines whether an error is present and where regeneration should begin.

Continued repair needs independent feedback. Useful repairs change as the student changes. Training-error feedback improves fit to training data while benchmark accuracy declines, whereas the fixed feedback set improves benchmark accuracy across rounds (Fig. 5c,d). Student failures are therefore best used as repair candidates, while **an independent feedback set identifies which capability gaps to target**; medical tags and error types make these gaps actionable.

Answer correctness and grounding are complementary. The human audit confirms high answer correctness and shows that visual grounding is a distinct, graded dimension of repair quality (§ 6.1). Medical self-evolution therefore benefits from assessing **evidence use alongside accuracy** and hint non-leakage.

6.4 Limitations and Future Directions

The current implementation has a fixed knowledge structure. Its taxonomy and error codebook constrain which weaknesses can be represented, and a strong proprietary auditor supplies the diagnoses. Evaluating open-source auditors and separating the effects of trajectory length from question difficulty would clarify these dependencies. Our benchmarks use multiple-choice and short-answer formats; clinical deployment requires **evaluation in the intended care setting**.

From diagnosis to grounded recovery. Future work will investigate **evidence-grounded recovery** through source-attributed retrieval, factuality checks, and image-region supervision. Evaluation should jointly measure answer leakage, grounding, and downstream learning.

Long-horizon and adaptive navigation. A research direction is **knowledge-structure evolution**: using persistent failures to propose expert-validated updates to the taxonomy and recovery operations. Longer runs can measure regression, cost, and saturation on independent evaluation data.

Potential for transferable learning strategies. Future studies can test the reuse of repair experience across modalities, specialties, and institutions, and its role in allocating annotation and auditor budgets. Key outcomes include transfer performance, rare-condition coverage, and robustness under distribution shift.

7 Conclusion

We introduced MedEvo and defined **Knowledge Navigation** as a feedback-guided process that uses structured domain knowledge and failure diagnosis to choose where to intervene, how to recover, and what to learn next. A gold-blind auditor localizes the first consequential error and provides a minimal, answer-free hint. The same student preserves its valid prefix and regenerates the erroneous suffix, and feedback on a fixed set directs subsequent repairs toward remaining weaknesses. **With SFT alone**, MedEvo improves Qwen3.5-9B, Qwen3.5-35B, and Qwen3.5-397B across seven medical multimodal benchmarks, and on Qwen3.5-9B these gains are reproducible across independent training runs.

Our analyses establish three findings. First, **failed trajectories contain reasoning worth preserving**. Second, **repair success and learning value are distinct objectives**. Third, **independent feedback is essential for sustained improvement**. Together, these findings identify repair selection and feedback-guided targeting as central to effective medical self-evolution.

Knowledge Navigation links error diagnosis to learning strategy. Future work can extend this framework through evidence-grounded repair, longer training horizons, and validated updates to the knowledge structure.

Declaration of Artificial Intelligence (AI) Assistance. AI tools were used solely for language editing, manuscript formatting, and figure preparation. They were not involved in method development or experimental design. All AI-assisted content was reviewed and revised by the authors, who take full responsibility for the final manuscript.

Reproducibility Statement. The method and experimental procedures are described in the main text and appendices. We will release the implementation, prompts, training configurations, evaluation scripts, and data-processing instructions to support reproducibility, subject to the licenses and access restrictions of the underlying datasets.

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A Method Details

This appendix provides implementation details that complement §3. Within round t , **the generator** $M_t = \pi_{\theta_t}$ **is frozen**: it produces rollouts, entropy profiles, continuations, and learnability scores, while the next student is trained from M_0 (Eq. 17).

A.1 Failure Definition and Step Segmentation

Given a question $x = (I, q, O)$ with reference answer $y \in O$, M_t samples a trajectory r and a parsed answer $\hat{y}$. A trajectory r^- fails if $\hat{y} \neq y$ or no valid answer can be parsed. Each failed trajectory is segmented into K **semantically complete steps** (Eq. 2), typically covering image observation, feature interpretation, mechanism, differential diagnosis, and option mapping. A boundary $b \in \{0, \dots, K-1\}$ splits r^- into the preserved prefix $p_b = (s_1, \dots, s_b)$ and the erroneous suffix $s^- = (s_{b+1}, \dots, s_K)$ (Eq. 8), where $b = 0$ denotes an empty prefix.

A.2 Entropy-Based Localization

Step scores. Token entropy H_j , the step score E_k^* , and the local surge ΔE_k follow Eqs. 3–5, with entropy measured in nats. We use $\lambda = 0.5$ and $m = 3$, with $m_k = \min(m, k-1)$ for $k > 1$ and $\Delta E_1 = 0$. Combining the mean with the 90th percentile $Q_{0.9}$ balances average uncertainty with an upper-tail signal within each step.

Anchor and window. Both profiles are **min–max normalized** within each trajectory, giving $\tilde{E}_k^*$ and $\tilde{\Delta E}_k$; a constant profile maps to zero. The anchor is $k^* \in \arg \max_{1 \leq k \leq K} (\tilde{E}_k^* + \tilde{\Delta E}_k)$, and the audit window is

$$\mathcal{W}(k^*) = \{s_{\max(1, k^* - w_\ell)}, \dots, s_{\min(K, k^* + w_r)}\}, \quad w_\ell = w_r = 1. \quad (18)$$

The step preceding the window defines a provisional boundary

$$b_0 = \max(0, k^* - w_\ell - 1). \quad (19)$$

Both $\mathcal{W}(k^*)$ and b_0 are proposals; **the auditor sets the final boundary** b (Appendix A.3).

Proposal assessment. For every audited trajectory, the auditor classifies the entropy proposal as HIT (the first consequential error lies at or adjacent to k^*), PARTIAL_HIT (the window covers the propagated error, but the unsupported premise begins earlier), MISS (the first error lies outside the window), or NON_ERROR_UNCERTAINTY (the high-entropy region is medically and logically valid). These labels are logged with each repair record and support the analysis in Fig. 6c.

A.3 Gold-Blind Audit and Recovery Operators

Auditor. We use GPT-5.6-sol as the auditor $\mathcal{A}$ in Eq. 7. Its inputs are the images, the question and options, the failed trajectory, the entropy profile (E_k^* , normalized scores, ΔE_k , k^* , and $\mathcal{W}(k^*)$), and **observable tags with provenance**. The reference answer, reference rationale, answer-derived tags, and retrieval metadata are **excluded through allowlist serialization** (Appendix C.1.3). The auditor returns a structured record (Appendix D.3) containing an action in {REPAIR, SKIP}, the first consequential error e with its type and subtype, the boundary b with a verbatim ending span (from s_b if $b > 0$, empty if $b = 0$), and the hint h . It returns SKIP when only answer formatting or option selection is wrong and no reasoning revision is justified. The final boundary may differ from b_0 . The auditor **traces evidence dependencies backward** and moves the boundary earlier when a later entropy surge stems from an upstream unsupported premise; conversely, it retains a high-entropy step that is medically and logically valid.

Recovery operators. An error type selects a recovery operator o ; the text hint h instantiates that operator in natural language. The operator belongs to

$$o \in \{\mathcal{P}_{\text{inspect}}, \mathcal{R}_{\text{reconstruct}}, \mathcal{D}_{\text{disambiguate}}, \mathcal{K}_{\text{bridge}}\}, \quad (20)$$

each paired with a parent error type (§3.1.2); subtype-level operations are listed in Appendix C.1.2. Operators specify the cognitive operation required for recovery.

- **Perceptual re-examination** ($\mathcal{P}_{\text{inspect}}$) names a visual region or attribute to revisit, such as lesion distribution, boundary shape, waveform morphology, stain localization, or cross-view consistency, without stating the finding.
- **Inference reconstruction** ($\mathcal{R}_{\text{reconstruct}}$) asks the student to rebuild the chain from evidence to mechanism, diagnosis, and the requested decision, for example when supportive and causal treatments are confused or care steps are misordered.
- **Differential disambiguation** ($\mathcal{D}_{\text{disambiguate}}$) requests comparison along the most discriminative feature, such as contrast pattern, onset profile, lesion location, signature pathology, or an exclusion criterion, without naming the diagnosis.
- **Knowledge bridging** ($\mathcal{K}_{\text{bridge}}$) supplies a narrowly scoped concept, such as a diagnostic criterion, drug mechanism, or immunohistochemical localization rule. It may state the concept but must not map it to an answer option.

Rendering. The cue $g(h)$ consists of a natural transition followed by the hint, with flexible wording. Error labels and audit explanations are **stored as metadata** and are never shown to the student.

A.4 Continuation, Admission, and Training Targets

Continuation budget. The current checkpoint generates eight rollouts per selected question. Each failed trajectory selected for repair is audited once and receives a single hint. With $p_b \oplus g(h)$ fixed, up to five suffixes are sampled (Eq. 9); **sampling stops at the first admitted continuation**. The user message contains x , the unfinished assistant message contains $p_b \oplus g(h)$, and generation continues within the same message **without repeating or modifying the prefix** (Appendix D.4).

Learnability gate. Using the suffix scores D_{repair} and Δ_{repair} (Eqs. 10–11), computed under M_t **without the hint**,

$$L_\xi = 1 \left[D_{\min} \leq D_{\text{repair}}(\xi) \leq D_{\max} \wedge \Delta_{\min} \leq \Delta_{\text{repair}}(\xi) \leq \Delta_{\max} \right], \quad (21)$$

with $[D_{\min}, D_{\max}] = [0.35, 0.85]$ and $[\Delta_{\min}, \Delta_{\max}] = [0.02, 0.30]$. The lower bound on Δ_{repair} requires that the student still prefers its original wrong suffix, so the repair carries information the student has not yet acquired. The upper bound limits the increase in per-token loss relative to the wrong suffix. Thresholds are selected on the feedback set and **frozen before final evaluation**. Feedback-set questions are **excluded from training targets**. Thresholds, retained fractions, and generator hashes are logged with each repair record (Appendix C.1.4).

Admission checks. Admission follows Eq. 12. C_ξ compares the parsed answer with the reference label. A_ξ **verifies that the exported prefix matches** the auditor’s verbatim span and that b precedes the diagnosed error. N_ξ **rejects continuations** that introduce symptoms, history, measurements, or findings absent from the input, as well as rep-

etition loops. Candidates with either suffix empty, nonfinite losses, or a nonpositive denominator in Eq. 10 are rejected.

Training targets and fallback. Admitted repairs are exported as $r^* = p_b \oplus g(h) \oplus s^+$ and trained with Eq. 13; original correct rollouts and historical repairs use the same **full-response loss**, and image and user tokens are masked. If no repair is admitted for a question, a quality-screened correct rollout or the existing valid target is retained; if neither exists, the question contributes no target. **No hint or auditor output is used at evaluation.**

A.5 Feedback Statistics, Quota Allocation, and Dataset Update

Feedback statistics. Besides the error rate in Eq. 14, consecutive checkpoints yield transition counts

$$N_{g,t}^{uv} = \sum_{i \in V_{\text{feedback},g}} \mathbb{1}[c_{i,t-1} = u \wedge c_{i,t} = v], \quad u, v \in \{0, 1\}, \quad t \geq 1, \quad (22)$$

which separate **corrected errors** (N^{01}), **regressions** (N^{10}), **persistent errors** (N^{00}), and **retained successes** (N^{11}). The priority in Eq. 15 normalizes regressions by n_g . For diagnostics, we also report the conditional regression rate $N_{g,t}^{10}/(N_{g,t}^{10} + N_{g,t}^{11})$ only when its denominator is positive; it is undefined and omitted when the preceding checkpoint has no correct examples. Groups comprise supported single-axis tags and predeclared intersections. A group with $n_g < n_{\min}$ backs off to its nearest supported ancestor, and identical resolved nodes are merged. Each question is counted once within a group, and overall statistics are computed over the deduplicated union of questions.

Quota allocation. From the second round on, the budget B_t is allocated in proportion to $S_{g,t}$ (Eq. 15), using largest-remainder rounding with deterministic tie-breaking. Quotas are **capped by the available unique training questions**, and surplus capacity is **redistributed among eligible groups**. Each multi-tag question is charged to the budget once. If all scores are zero, questions are sampled uniformly from eligible unique questions. Token and auditor-call caps impose additional limits, and any unfilled quota is recorded. The priority and quota rule is Eq. 15; the planning template in Appendix D.2 assembles a plan and tag-based retrieval requests subject to these numerical constraints. The validator in Appendix C.1.4 checks the plan before execution.

Retrieval and repair priority. Within each group, candidates are retrieved by tag match and ranked by historical error records. Candidates may include questions already used in training. Selected questions receive **fresh rollouts** from M_t , and their failures are diagnosed anew. Failures matching both the modality and the error type prioritized by feedback are repaired first, followed by modality-only or error-type-only matches. Matching uses **current diagnoses**. Priority determines where repair effort is spent; admission (Appendix A.4) determines what enters training.

Dataset update. Questions are deduplicated by canonical ID, and each keeps **at most one active target**. An admitted candidate replaces the existing target only if a comparison auditor, blinded to round and model labels, judges it better in evidence support, reasoning consistency, and conciseness; **ties retain the existing target**. All versions and rollout-specific diagnoses are logged in a separate history. Each round retrains from M_0 on the complete updated dataset (Eqs. 16–17) with a fresh optimizer and learning-rate schedule. Generation caps are fixed in advance, and plan records follow Appendix C.1.4.

A.6 Settings of Analysis Experiments

Unless stated otherwise, analyses use Qwen3.5-9B with the training configuration in Table 14 and the evaluation protocol in §4.1.

Repair designs (Table 4). All variants are applied in the first round under the same rollout and continuation budgets. Hint + full rewrite uses the type-guided hint but regenerates the trajectory from the start of the response. No-hint rewrite regenerates the full trajectory without a hint. Generic hint + suffix continuation keeps the auditor’s boundary but replaces the type-guided hint with a fixed generic cue.

Feedback ablations (Table 5). All variants start from the shared R1 checkpoint and use the same budgets. No feedback omits feedback-guided question selection. Without medical tags omits medical-tag feedback. Without error-type feedback uses tag-group quotas but does not match repairs to prioritized error types.

Trajectory selection (Fig. 5a). The longest-8k subset contains the 8,000 questions whose failed trajectories are longest; the random-8k subset is sampled uniformly. Each subset is used for one round of continuation training from M_0 .

Hint count (Fig. 5b). On the longest-8k subset, each failed trajectory receives $k \in \{1, 3, 5, 8\}$ hints, with five continuation attempts.

Feedback source (Fig. 5c,d). Training-error feedback uses errors from the training data to guide selection instead of errors on the feedback set.

Retry budget and entropy (Fig. 6b). We vary the continuation budget over $\{1, 3, 5, 8\}$ with and without entropy localization. Final acceptance is the fraction of repair candidates admitted, computed over 45,000 attempts per setting with 95% Wilson intervals.

Entropy bins and difficulty (Fig. 6c,d). The analysis compares erroneous-step rates across entropy bins. Difficulty levels follow the rubric in Appendix C.1.1.

B Dataset Statistics

B.1 Dataset Roles and Evaluation Scope

The study separates the training bank, the fixed feedback set, and the final evaluation benchmarks. Their roles differ: training questions supply supervision, feedback questions guide diagnosis and data selection, and benchmark questions measure downstream performance.

Table 9 Dataset roles and reported sizes. Test coverage is reported as the number of benchmarks.

Resource	Size	Role
Training bank	81,293 questions	Source of original correct rollouts, accepted repairs, and retained valid targets.
Fixed feedback set	5,000 questions	Medical capability diagnosis and subsequent data selection; excluded from training.
Final evaluation	7 benchmarks	Independent downstream accuracy reporting.

The training bank is constructed from public medical training splits and deduplicated using question and source identifiers, available patient or case identifiers, image hashes, and near-duplicate clusters. See Appendix C.1.4 for structured records and plan validation, and Appendix C.1.3 for information-access rules. Fig. 3 shows nine modality categories and five difficulty levels; the tag hierarchy is described in Appendix C.1.1.

Evaluation uses MedUniEval from Hulu-Med [25], built on Lingshu’s MedEvalKit [11]. The seven benchmarks are PMC-VQA [26], VQA-RAD [27], SLAKE [28], PathVQA [29], OmniMedVQA [30], MedXpertQA-MM [4], and the medical subset of MMMU [31]. Their accuracies are averaged without weighting by benchmark size. Modality and error-family results below describe the feedback set, separately from benchmark evaluation.

B.2 Round-wise Data Construction

Table 1 highlights a declining repair workload across training rounds. The number of questions submitted for repair falls from **27,233 to 23,620**, a reduction of **3,613 questions (13.27%)** relative to R1. Their share of the fixed question bank decreases from 33.50% to 29.06%. The unit of this workload measure is a question submitted to the repair stage.

The final training set remains fixed at **81,293 questions** in each round, combining qualified original correct trajectories with **accepted repairs** and retained valid historical targets. Thus, repair workload decreases while training coverage stays constant. The number of successfully repaired questions decreases from 19,567 to 16,797 as fewer questions are submitted. Repair success rates are 71.85%, 70.68%, and 71.11%, remaining stable across rounds. With at most five continuation attempts per repair, the mean number of attempts also falls from **3.37 to 2.12**, indicating improved hint-guided correction efficiency. Each column identifies the generator checkpoint and the subsequent training round.

B.3 Additional Experimental Details

B.3.1 Evolution across Twenty Error Subtypes

Fig. 7 expands the four families into twenty subtypes. Bar heights are normalized by each subtype’s R0 rate, while in-bar labels retain the reported percentages. All subtypes are below R0 at R3. The knowledge-gap family has an R3/R0 ratio of approximately 0.89, compared with 0.72–0.75 for the other families, indicating a smaller relative reduction in the reported breakdown, conditional on a consistent counting denominator across rounds. Rates are reported separately by error subtype.

B.3.2 Round-wise Feedback Statistics

Tables 10 and 11 report the per-round modality error rates and error-family percentages; §4.3 summarizes the modality trends. Blue downward arrows indicate decreases, red upward arrows increases, and equality signs unchanged rates. The error-family percentages are reported separately from the modality-level error rates.

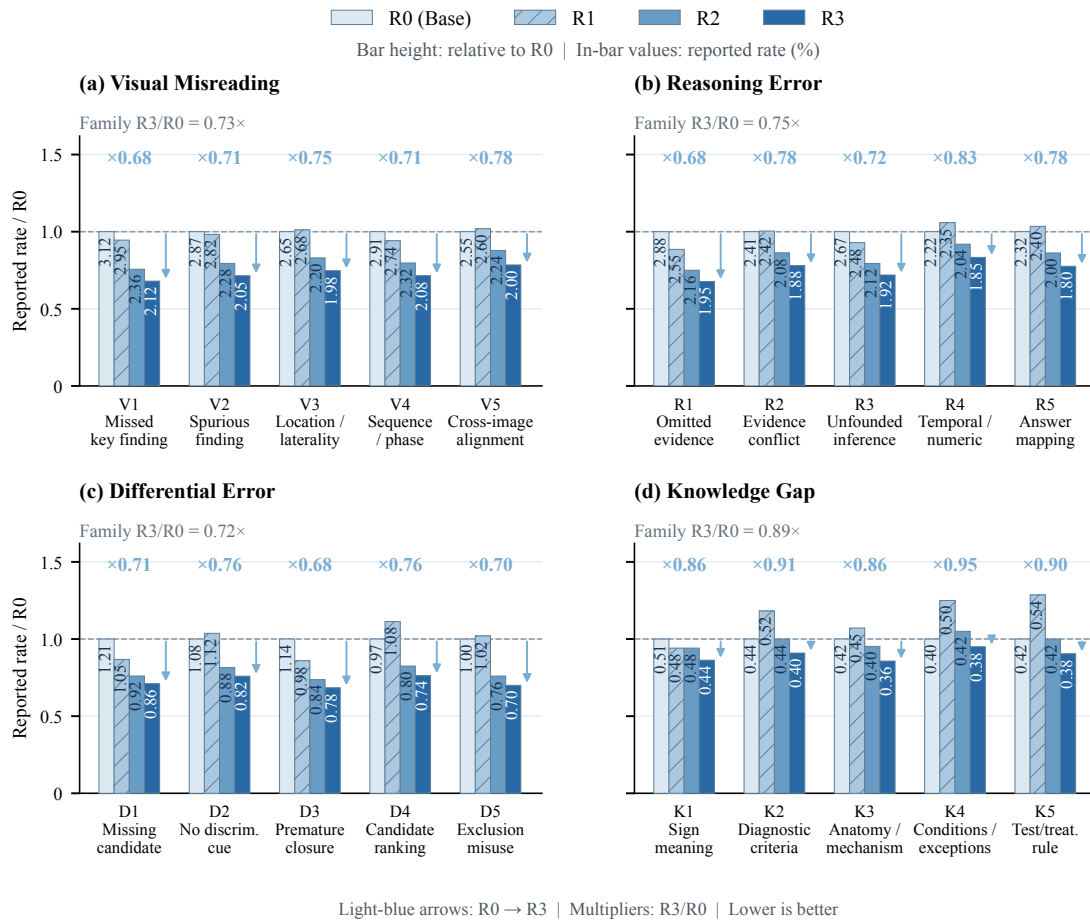


Figure 7 Twenty error subtypes of the 9B model across R0–R3. Each panel contains five subtypes. Bar heights divide reported rates by the corresponding subtype’s R0 rate; in-bar labels give original percentages. Light-blue arrows and multipliers show R3 relative to R0, with lower values indicating larger relative reductions. Family R3/R0 is the ratio of the summed subtype rates. Horizontal codes such as R1–R5 denote reasoning-error subtypes, distinct from round labels R0–R3 in the legend.

Table 10 Modality error rates (%) on the fixed feedback set; N is group size. Blue shading marks R3. Arrows compare consecutive rounds; bold/underlining mark the two lowest rates.

Modality	N	R0	R1	R2	R3
Radiology	1509	37.18	35.98	↓ 31.28	↓ <u>32.21</u> ↑
Nuclear medicine	234	34.62	26.50	↓ <u>24.79</u>	↓ 22.65 ↓
Ultrasound	351	44.73	39.03	↓ 39.60 ↑	↑ 36.18 ↓
Endoscopy	247	32.39	30.36	↓ <u>27.13</u>	↓ 21.86 ↓
Pathology	378	34.13	34.66	↑ <u>30.95</u>	↓ 28.31 ↓
Ophthalmology	135	34.07	27.41	↓ <u>22.96</u>	↓ 20.00 ↓
Dermatology	593	30.35	27.82	↓ <u>24.62</u>	↓ 23.10 ↓
Dental imaging	308	40.26	37.01	↓ 37.66 ↑	↑ 32.79 ↓
Functional tests	1245	20.00	17.75	↓ <u>16.22</u>	↓ 15.50 ↓

Table 11 Error-family percentages, summed over five subtypes per family. These values are not directly comparable to the modality error rates. Blue shading marks R3; arrows and bold/underlining follow Table 10.

Error family	R0	R1	R2	R3
Visual misreading	14.10	13.79	↓ <u>11.40</u>	↓ 10.23 ↓
Reasoning error	12.50	12.20	↓ <u>10.40</u>	↓ 9.40 ↓
Differential error	5.40	5.25	↓ <u>4.20</u>	↓ 3.90 ↓
Knowledge gap	2.19	2.49	↑ <u>2.16</u>	↓ 1.96 ↓

B.4 Detailed Error Statistics for the Current Rounds

Table 12 Error-subtype rates (%): visual/reasoning subtypes (left) and differential/knowledge subtypes (right). Blue shading marks R3. Subtype codes R1–R5 denote reasoning errors, not training rounds. Arrows compare each rate with the preceding round: blue ↓ marks a decrease, red ↑ an increase, and = no change. Bold and underlining mark the lowest and second-lowest distinct rates in each row.

Subtype	R0	R1	R2	R3	Subtype	R0	R1	R2	R3
V1 Missed key finding	3.12	2.95	↓ 2.36	↓ 2.12	D1 Missing candidate	1.21	1.05	↓ 0.92	↓ 0.86
V2 Spurious finding	2.87	2.82	↓ 2.28	↓ 2.05	D2 No discriminating cue	1.08	1.12	↑ 0.88	↓ 0.82
V3 Location / laterality	2.65	2.68	↑ 2.20	↓ 1.98	D3 Premature closure	1.14	0.98	↓ 0.84	↓ 0.78
V4 Sequence / phase	2.91	2.74	↓ 2.32	↓ 2.08	D4 Candidate ranking	0.97	1.08	↑ 0.80	↓ 0.74
V5 Cross-image alignment	2.55	2.60	↑ 2.24	↓ 2.00	D5 Exclusion misuse	1.00	1.02	↑ 0.76	↓ 0.70
R1 Omitted evidence	2.88	2.55	↓ 2.16	↓ 1.95	K1 Sign interpretation	0.51	0.48	↓ 0.48	= 0.44
R2 Conflicting evidence	2.41	2.42	↑ 2.08	↓ 1.88	K2 Diagnostic criteria	0.44	0.52	↑ 0.44	↓ 0.40
R3 Unsupported inference	2.67	2.48	↓ 2.12	↓ 1.92	K3 Anatomy / mechanism	0.42	0.45	↑ 0.40	↓ 0.36
R4 Temporal / numeric	2.22	2.35	↑ 2.04	↓ 1.85	K4 Conditions / exceptions	0.40	0.50	↑ 0.42	↓ 0.38
R5 Answer mapping	2.32	2.40	↑ 2.00	↓ 1.80	K5 Test / treatment rule	0.42	0.54	↑ 0.42	↓ 0.38

Table 13 separates repair volume from success rate. Counts measure attempted and successful repairs within each error family.

Reasoning-error repair success rises from 56.43% at R1 to 83.18% at R3, while knowledge-gap repair success falls from 90.28% to 48.82%. These rates describe the repair candidates submitted in each round.

Table 13 Repair counts and success rates by error family. Gray arrows track counts; blue/red arrows mark rising/falling success rates. Bold/underlining mark best/second-best rates per family; blue shading marks R3.

Error family	Round	Attempted	Successful	Success (%)
Visual misreading	R1	12,189	9,826	80.61
	R2	10,245 ↓	6,789 ↓	66.27 ↓
	R3	10,478 ↑	7,094 ↑	<u>67.70</u> ↑
Reasoning error	R1	9,355	5,279	56.43
	R2	7,857 ↓	6,246 ↑	<u>79.50</u> ↑
	R3	6,789 ↓	5,647 ↓	83.18 ↑
Differential error	R1	4,064	2,995	73.70
	R2	6,742 ↑	4,673 ↑	<u>69.31</u> ↓
	R3	4,786 ↓	3,291 ↓	68.76 ↓
Knowledge gap	R1	1,625	1,467	90.28
	R2	987 ↓	550 ↓	<u>55.72</u> ↓
	R3	1,567 ↑	765 ↑	48.82 ↓

C Medical Taxonomy and Error Categories

C.1 Taxonomy, Permissions, and Execution Contracts

C.1.1 Four-Dimensional Medical Tag Dictionary

The medical taxonomy describes each question along four independent axes: modality, anatomy, disease, and difficulty. Each axis allows multiple hierarchical labels for capability analysis and question retrieval.

Modality hierarchy. Computed tomography (CT), magnetic resonance imaging (MRI), positron emission tomography (PET), and single-photon emission computed tomography (SPECT) denote imaging modalities. Other abbreviations used below are optical coherence tomography (OCT), fundus fluorescein angiography (FFA), indocyanine green angiography (ICGA), cone-beam computed tomography (CBCT), electroencephalogram (EEG), and electromyogram (EMG). ECG denotes electrocardiogram, as in Fig. 3.

Parent	Second-level scope
1 Radiography	X-ray; CT; MRI; mammography; fluoroscopy; angiography
2 Nuclear medicine	PET/CT; SPECT; radionuclide scans
3 Ultrasound	B-mode; echocardiography; vascular; interventional
4 Endoscopy	Gastrointestinal; bronchial; laryngeal/nasal; bladder; joint; laparoscopic
5 Pathology	Histology; cytology; molecular pathology
6 Ophthalmology	Fundus photography; OCT; FFA/ICGA; slit lamp
7 Dermatology	Dermoscopy; clinical photography; reflectance confocal microscopy
8 Dental	Panoramic; CBCT; periapical; cephalometric
9 Functional tests	ECG; EEG; EMG

Anatomy hierarchy.

Parent	Second-level scope
1 Nervous system	Brain; spinal cord; peripheral nerves
2 Head and neck	Orbit; ear; sinuses; oral/maxillofacial; pharynx/larynx; neck soft tissue
3 Thorax	Lung/airway; mediastinum; pleura; heart; great vessels; breast; chest wall
4 Abdomen	Hepatobiliary; pancreas; spleen; gastrointestinal; renal/urinary; retroperitoneum
5 Pelvis	Male reproductive; female reproductive; pelvic floor; pelvic bones
6 Spine	Cervical; thoracic; lumbar; sacrococcygeal; disc; canal; paraspinal tissue
7 Musculoskeletal	Craniofacial bones; upper limb; lower limb; muscles; periarticular soft tissue
8 Vascular	Arteries; veins; lymphatic system
9 Skin and soft tissue	Head/face; trunk; limbs; subcutaneous; deep soft tissue

Disease hierarchy.

Parent	Second-level scope
1 Neoplasm	Malignant; benign; metastatic; precancerous categories
2 Infection and inflammation	Bacterial; tuberculosis; viral; fungal; parasitic; noninfectious inflammation
3 Vascular disease	Ischemic; hemorrhagic; aneurysm; thrombosis/embolism; malformations; atherosclerosis/stenosis
4 Degenerative disease	Joint; neural; organ degeneration
5 Trauma	Fracture; joint injury; visceral injury; head injury; foreign body
6 Congenital/developmental	Cardiovascular; nervous; skeletal; digestive; urinary; respiratory
7 Metabolic/endocrine	Diabetes complications; thyroid; adrenal; gout; metabolic bone
8 Autoimmune	Lupus; ankylosing spondylitis; multiple sclerosis; neuromyelitis optica spectrum disorder (NMOSD); autoimmune liver; immunoglobulin G4 (IgG4); vasculitis; systemic sclerosis; myositis; Sjogren
9 Functional disorders	Obstruction; effusion; ectopic gas; torsion; stones
10 Iatrogenic/postoperative	Normal postoperative change; complications; implants; post-radiotherapy; drug-related

Leaf specificity and multi-label assignment. CT branches into noncontrast, contrast-enhanced, CT angiography (CTA), high-resolution CT (HRCT), and low-dose CT; MRI includes T1- and T2-weighted imaging, fluid-attenuated inversion recovery (FLAIR), diffusion-weighted imaging (DWI), magnetic resonance angiography (MRA), functional MRI (fMRI), magnetic resonance spectroscopy (MRS), and susceptibility-weighted imaging (SWI). Anatomy can descend from thorax to lung/airway and then to a lung lobe, or from abdomen to pancreas and then head/body/tail. Disease labels descend from a broad family to a mechanism or disease group and then to a named entry. Preserve image IDs for mixed modalities and record every supported anatomical site. **Assign the deepest supported node**, not a guessed leaf; use unknown, not-applicable, or a supported parent when necessary. **Expand ancestors as a set** and count each question once per group. Related concepts at different paths require an explicit alias ledger, not automatic merging. For each label, store source text, evidence span/image ID, confidence, visibility, and taxonomy version.

Difficulty rubric.

Level	Task requirement
L1	Single modality and conspicuous canonical findings; direct recognition.
L2	Common presentation requiring localization or characterization and limited context.
L3	Atypical or multiple findings requiring sequence integration or differential comparison.
L4	Subtle findings, multisystem involvement, or complex context requiring extensive integration.
L5	Highly complex or disputed tasks requiring multidisciplinary, quantitative, or specialized knowledge.

Assign difficulty from the actual question: visibility, number of required comparisons, evidence integration, and specialized knowledge. A disease name, modality, or a single failed rollout does not determine the level. Rare disease alone does not force L5. Record the rationale and adjudicate ambiguous adjacent levels.

C.1.2 Error Codebook: Four Parents and Twenty Subtypes

We specify five subtypes under each of the four existing parent categories. Each row defines an observable diagnostic criterion and a navigation operation; it does not supply a gold answer.

V: Visual misreading.

Code / subtype	Diagnostic criterion	Navigation operation
V1 Missed finding	A supported salient finding is not observed or described.	Reinspect the relevant region.
V2 Hallucinated or misidentified finding	An artifact, normal structure, or unsupported feature is read as a lesion.	Check direct visual support.
V3 Anatomy or laterality error	The organ, level, compartment, or side conflicts with visible evidence.	Recheck landmarks and orientation.
V4 Modality, sequence, or phase error	Acquisition type, sequence, or phase is misidentified, distorting feature reading.	Verify acquisition context before reading features.
V5 Cross-image correspondence error	A lesion is matched incorrectly across images, slices, or views.	Align images and anatomical locations.

R: Reasoning error.

Code / subtype	Diagnostic criterion	Navigation operation
R1 Omitted evidence	An observed finding or stated fact is omitted from inference.	Reintroduce the omitted evidence.
R2 Conflicting evidence integration	Correctly extracted evidence conflicts and is not reconciled.	Reconcile supporting and opposing evidence.
R3 Unsupported inference or causal leap	A conclusion lacks the required logical or causal link.	Restore a testable intermediate inference.
R4 Temporal or quantitative error	Correctly read facts are combined with an incorrect temporal or numerical relation.	Check chronology, units, or calculation.
R5 Task constraint or answer-mapping error	The reasoning violates the requested target, negation, or option mapping.	Recheck the requested target and answer mapping.

D: Wrong differential.

Code / subtype	Diagnostic criterion	Navigation operation
D1 Omitted candidate	A necessary evidence-compatible alternative is not considered.	Broaden the candidate comparison.
D2 Missing discriminative comparison	Alternatives are listed without comparing discriminative evidence.	Identify discriminative evidence.
D3 Premature closure or anchoring	An initial diagnosis is fixed before counterevidence is considered.	Revisit counterevidence and alternatives.
D4 Incorrect candidate ranking	Known support, counterevidence, or priors are weighted incorrectly.	Compare the strength of evidence.
D5 Misapplied exclusion	An alternative is excluded using insufficient negative evidence or a nonessential criterion.	Check whether exclusion is justified.

K: Knowledge gap.

Code / subtype	Diagnostic criterion	Navigation operation
K1 Missing finding interpretation	The required medical meaning of a finding or term is missing.	Cue the relevant conceptual relation.
K2 Missing diagnostic criterion	A required diagnostic, grading, or staging rule is unavailable.	Cue the relevant rule category.

Code / subtype	Diagnostic criterion	Navigation operation
K3 Missing anatomy or pathophysiology	Knowledge linking anatomy, mechanism, and manifestations is missing.	Bridge the minimal mechanistic relation.
K4 Missing applicability condition or exception	A general rule is known but its prerequisites or exceptions are missing.	Cue the applicability check.
K5 Missing investigation or management knowledge	A next-investigation or management task requires an unavailable principle.	Cue the decision principle without the answer.

Boundary rules and reporting. A finding not perceived is V1; a finding already perceived but omitted from inference is R1. Incorrect correspondence across images is V5; conflicting correctly extracted evidence is R2. Candidate-set construction, comparison, and exclusion belong to D; a demonstrated missing prerequisite rule belongs to K. Knowledge-gap labels require evidence of a missing prerequisite rule. **Select the earliest consequential supported error** as the primary subtype and record later failures as secondary labels, with step spans and evidence. If the parent is clear but the subtype is not, retain the parent and an unresolved-subtype status; if even the parent is unsupported, mark the audit unresolved. These are annotation states, not extra error classes. Use **one primary subtype per trajectory** for additive counts; report secondary labels separately. Aggregate V1–V5 into V, and similarly for R, D, and K. Re-audit fresh rollouts at each checkpoint using subtype-specific annotations and denominators.

C.1.3 Information Access Matrix

Component	Permitted inputs	Excluded inputs
Feedback evaluator	Feedback items, labels, current predictions; emits aggregates and transitions	Final-test results for planning; feedback export into SFT
Planner / retrieval	Aggregate feedback, training tag metadata (including true disease labels), coverage, budgets	Final-test items/results; any authority to forward answer-derived labels to repair
Auditor / hint renderer	Original images, question/options, current failed trajectory, entropy profile, observable tags with provenance	Gold answer, answer-derived disease/location, reference rationale, hidden retrieval labels
Continuation M_t	Original input and assistant prefill $p_b \oplus g(h)$	Gold answer, full audit, hidden tags, teacher solution
Admission / exporter	Candidate, reference answer for verification, boundary, provenance, suffix scores	Feedback/test examples as training targets
SFT trainer	Frozen training manifest; $p_b \oplus g(h) \oplus s^+$, masks	Full audit as model input; feedback/test samples

Use **explicit allowlist serialization**, not deletion of one answer field from an otherwise unrestricted object. **Reject fields without observable provenance**, including filenames/IDs encoding labels. An authorized evaluator can identify a wrong training answer before repair without transmitting the answer. Generation sees only a boolean routing decision, not the key. Record hashes of redacted requests for leakage audit. The planner’s chosen hint family is passed through an eligibility check derived from observable context and current diagnosis; disease-bearing planner explanations are never forwarded.

C.1.4 Structured Fields and Plan Validation

The execution interface requires JavaScript Object Notation (JSON) records containing the following fields. An omitted measurement is null with a reason; zero means measured zero.

Question metadata: canonical question ID; source ID; patient/case/article cluster IDs when available; image hashes and near-duplicate cluster; split; taxonomy version; per-image modality; all four tag sets; per-tag provenance, visibility (statistical or observable), and uncertainty.

Feedback record: checkpoint hash; split hash; group and resolved parent; unique sample count; error count/rate; all four transition counts; both regression denominators; current error-type/subtype counts; confidence interval; training-pool unique availability; measured repair/admission/cost fields.

Plan: round ID; generator and initialization hashes; planner type/version and seed; feedback hash; target group IDs; numeric feedback evidence and fallback ancestry; training sources; selected IDs; attempt/call/token ceilings; per-group quota; permissible hint operation; history replay allocation; replacement rule; supervised-token cap; shortfall/reallocation policy; **fixed training-config hash**; acceptance-validator result.

Repair record: question and checkpoint IDs; rollout hash; step and token spans; entropy profile and proposal;

current error type/subtype; audit boundary and exact prefix span; observable-tag payload hash; recovery cue; each continuation and seed; correctness and integrity checks; current-model suffix scores and cutoffs; admission/rejection reason; all generated/scored tokens.

Mix and training record: ordered example IDs; source round and generator; dedup/replacement ledger; historical/new role; token masks; sequence lengths; unit weights; packing map; effective repetitions and exposures; optimizer and scheduler reset flags; learning rate/warmup; initialization hash; output hash; second-stage and cumulative costs.

Execution receipt: planned and actual quotas, unique selected and attempted failures, successful and admitted repairs, substitutions, exhausted budgets, rejected records, uncovered groups, and actual training tokens.

The validator rejects cross-split IDs, duplicate quota charges, budgets above caps, out-of-pool retrieval, unsupported hint operations, missing visibility provenance, or changed training hyperparameters. Save the validated plan before generation and record actual execution counts afterward.

The main-agent feedback and next-round planning prompt is given in Appendix D.2; the trajectory audit prompt is given in Appendix D.3.

C.1.5 Training Configuration and Evaluation Scope

Training uses brain floating-point 16 (BF16) precision on graphics processing units (GPUs). Parallelism abbreviations are tensor parallelism (TP), expert parallelism (EP), pipeline parallelism (PP), context parallelism (CP), expert tensor parallelism (ETP), and data parallelism (DP); the following number gives the parallelism degree. Configuration keys retain their literal software spelling: llm denotes large language model, vit denotes vision transformer, cpu denotes central processing unit, moe denotes mixture of experts, gemm denotes general matrix multiplication, mlp denotes multilayer perceptron, and lr denotes learning rate.

Table 14 Recorded MoE training configuration.

Configuration	Value	Configuration	Value
Framework	ms-swift + Megatron-LM backend	add_non_thinking_prefix	true
Precision	BF16	loss_scale	ignore_empty_think
micro_batch_size	1	MAX_PIXELS	3145728
global_batch_size	64	dataset_num_proc	96
packing	true	dataloader_num_workers	8
attention_backend	flash	logging_steps	1
calculate_per_token_loss	true	eval_iters	1000
freeze_llm / freeze_vit / freeze_aligner	false / true / true	recompute_granularity	selective
optimizer_cpu_offload	true	sequence_parallel	true
use_precision_aware_optimizer	true	max_length	28672
offload_fraction	1.0	epochs	2
moe_grouped_gemm	true	save_total_limit	2
moe_shared_expert_overlap	true	Recompute modules	core_attn + moe + mlp
moe_aux_loss_coeff	1e-6	moe_permute_fusion	true
lr	1e-5	Parallelism	TP4/EP8/PP1/CP1/ETP1; 64 GPUs
lr_warmup_fraction	0.05		
min_lr	1e-7		

EP and MoE-specific options apply only to MoE models.

Both example types use full-response loss, retaining recovery cues in repairs.

§4 reports accuracy on seven benchmarks and their mean, comparing each backbone’s best complete checkpoint with Base.

D Prompts

D.1 Shared Question-Answering Prompt

All question-answering tasks use the following instruction verbatim.

Shared Question-Answering Instruction

Please reason step by step, and put your final answer within `\boxed{\}`.

D.2 Main-Agent Feedback and Next-Round Planning

The following template connects feedback diagnosis to next-round selection and repair priorities under the method protocol. It specifies agent behavior; run-specific inputs and execution records are required to document its use.

Main Agent: Feedback Summary and Next-Round Data Plan

You are the feedback and data-planning agent for MedEvo. Diagnose the current model’s weaknesses and produce an executable next-round data plan. Do not write solutions to training questions or change the experimental comparison settings.

Inputs. Current round and checkpoint; the fixed 5,000-question feedback-set results for current and previous checkpoints; four-axis medical tags; four error families and their twenty subtypes; per-group sample counts, correctness transitions, and current error diagnoses; available training-question IDs and tags; historical trajectory records; fixed generation/repair budgets, support threshold, regression weight α , and admission rules. Missing inputs must be reported, never invented.

1. Summarize feedback. For each supported tag group, report error rate, change from the previous round, corrected errors, regressions, and persistent errors. Summarize four-family and twenty-subtype distributions with explicit denominators. Count each question once within a group; do not sum overlapping groups into an overall rate. Merge unsupported groups into supported parents. Distinguish observed changes from explanations that require further evidence.

2. Set priorities. Rank groups using $S_{g,t} = \epsilon_{g,t} + \alpha N_{g,t}^{10}/n_g$ under the supplied configuration. Identify dominant or persistent error types within each priority group. State which measured feedback supports each priority; do not infer that a high-error modality contains every error subtype.

3. Select training questions. Allocate the fixed budget proportionally to group scores, cap quotas by available unique questions, and redistribute unused quota. Preserve modality coverage and apply the configured fallback when all scores are zero or support is insufficient. Retrieve by medical tags, using historical errors only as prioritization cues. Deduplicate multi-tag questions and return selected training IDs, group quotas, and any shortfall. **Never select feedback or final-test examples.**

4. Guide fresh rollouts and repair. Ask the current checkpoint to generate the configured eight rollouts per selected question. **Diagnose their actual errors again.** Use feedback-identified modality–error pairs to prioritize failed training trajectories: prefer joint matches, followed by modality-only or error-type-only matches when necessary. For each selected trajectory, pass its observable medical tags and current error diagnosis to the hint generator. The recovery hint must address the error actually observed in this trajectory. Send observable tags and local diagnoses to the auditor and hint generator, **without reference answers or answer-bearing metadata.** Retain correct rollouts and accepted continuations under the existing screening rules; if repair fails, use an available quality-screened correct rollout. If none is available, retain the valid historical target or report no accepted target.

5. Plan the next round. **Keep at most one active target per question** and replace it only with an accepted improvement. **Train on the complete updated dataset from M_0 .** Keep training settings and evaluation protocol fixed. Evaluate the next checkpoint on the same feedback set to measure the targeted changes; independent benchmark results are reported separately and must not drive selection.

Output. Return a structured plan containing: round and checkpoint; feedback summary and denominators; priority tag/error pairs and supporting statistics; score and quota per group; selected unique training IDs; rollout and repair budgets; repair priorities; dataset-retention/replacement rules; initialization M_0 ; next-round feedback comparisons; and missing inputs or budget shortfalls. After execution, record actual generated, repaired, admitted, and replaced counts.

D.3 Stage 1 Audit Prompt

The diagnostic model receives the image, the question with options, the failed trajectory, the complete step-level uncertainty profile, and the entropy-derived anchor and candidate window. The auditor is gold-blind: **the reference answer is not an input to diagnosis.** Answer-derived disease labels, hidden lesion positions, reference explanations, and planner retrieval metadata are also excluded. Only provenance-checked observable tags are allowed. The entropy proposal is a localization prior, not an error label: the auditor inspects the proposed region first, traces its evidence dependencies backward, and may move the final boundary when the medical or logical

evidence requires it. It returns a structured audit that identifies a compact revision region and constrains the subsequent continuation. The emitted hint must remain answer-blind. Placeholders are shown in braces and are filled at runtime.

Prompt D.3: Stage-1 Entropy-Guided Medical Trajectory Audit

You are a senior medical reviewer auditing a failed multimodal reasoning trajectory segmented into semantic steps $s_1, \dots, s_K$.

You are given:

- (1) the medical image(s);
- (2) the clinical question and answer options;
- (3) the failed student trajectory segmented into semantic steps;
- (4) STEP_ENTROPY: the uncertainty score E_k^* of every step;
- (5) NORMALIZED_ENTROPY: the normalized uncertainty score of every step;
- (6) LOCAL_SURGE: the local increase ΔE_k relative to the preceding step history;
- (7) ENTROPY_ANCHOR: the step proposed by the uncertainty locator;
- (8) ENTROPY_WINDOW: the compact candidate window surrounding that anchor, and ENTROPY_THRESHOLD if a threshold was used.
- (9) OBSERVABLE_TAGS: per-image modality and other context supported by the original input, with evidence provenance; never hidden answer-derived tags.

The entropy signal guides audit localization. High uncertainty can reflect a genuine failure, but it can also arise from difficult terminology, ambiguous visual evidence, or a medically valid differential diagnosis. Combine the entropy proposal with the image, clinical evidence, and dependencies between reasoning steps to identify the first consequential failure.

Error-conditioned hint construction. Construct the hint from OBSERVABLE_TAGS, ERROR_TYPE, ERROR_SUBTYPE, and KEY_ERROR. Select a recovery operation addressing the diagnosed failure:

- **Visual misreading:** direct attention to the relevant region, visual attribute, or correspondence across available images.
- **Reasoning error:** revisit an unsupported inference or reconnect observed evidence to the requested decision.
- **Differential diagnosis error:** compare discriminative evidence or reconsider an insufficiently examined alternative.
- **Knowledge gap:** identify the relevant concept, rule, or applicability condition to reconsider.

Specify one concrete next operation grounded in the current question and trajectory. Express it as a brief recovery cue without supplying the corrected finding, final diagnosis, or answer option.

Audit procedure:

- (1) Inspect ENTROPY_WINDOW first. Determine whether its local entropy peak or surge corresponds to an unsupported observation, invalid inference, incorrect differential comparison, or missing medical principle.
- (2) Trace evidence dependencies backward from ENTROPY_ANCHOR. Decide whether the decisive failure begins inside the proposed window or an earlier step introduced the unsupported premise that caused the later entropy increase.
- (3) Do not automatically select the highest-entropy step. Justify the final AUDIT_ANCHOR and SAFE_PREFIX_END using the medical image, clinical information, and logical dependencies.
- (4) Prefer a compact localized repair and preserve every preceding step that remains medically and logically valid.
- (5) You may move the audit anchor outside ENTROPY_WINDOW when the proposal misses the first consequential failure, but briefly explain why the entropy signal was misleading.

Return a JavaScript Object Notation (JSON) object with the following fields, grounded strictly in the provided image and text:

- (a) ACTION: REPAIR or SKIP.
- (b) STEP_SUMMARY: a brief one-sentence semantic summary for each step.

- (c) ENTROPY_ASSESSMENT, containing: proposed_anchor, proposed_window, proposal_status, and brief_reason. proposal_status must be one of {HIT, PARTIAL_HIT, MISS, NON_ERROR_UNCERTAINTY}. HIT means that the first consequential failure is at or immediately adjacent to the proposed anchor; PARTIAL_HIT means that the proposal covers the propagated error region but the unsupported premise begins earlier; MISS means that the first failure lies outside the proposed window; and NON_ERROR_UNCERTAINTY means that the high-entropy region is medically or logically valid.
- (d) AUDIT_ANCHOR: the index of the first consequentially incorrect step.
- (e) SAFE_PREFIX_END, containing step_index and the exact verbatim_ending_span of the last fully valid step before the repair region.
- (f) AUDIT_WINDOW: a short contiguous sequence of steps that should be reconsidered.
- (g) ERROR_TYPE and ERROR_SUBTYPE: checkpoint-specific diagnosis, plus KEY_ERROR: the localized error stated without revealing the correct answer.
- (h) HINT_TYPE: one of {perceptual_reexamination, inference_reconstruction, differential_disambiguation, knowledge_bridging}.
- (i) MINIMAL_HINT: a brief recovery cue with flexible wording, conditioned on observable medical tags and the diagnosed local error. Name the evidence or reasoning operation to revisit. Do not include the error-category label or audit explanation. Introduce the cue with a natural transition; no fixed phrase or first-person form is required.
- (j) IMAGE_CHECK, containing applicable (true or false) and target (the specific modality, anatomy, laterality, finding, relation, or measurement to re-inspect).
- (k) KNOWLEDGE_SCOPE: the narrow medical principle to recall, or null.
- (l) DIFFERENTIAL_FOCUS: the discriminative axis to compare, or null.
- (m) STOP_CONDITION: the evidence condition under which the continuation should commit to an answer and stop.

Constraints:

- Do not access hidden disease labels or hidden lesion coordinates.
- Do not state a gold answer, corrected diagnosis, or correct option letter.
- Do not provide a replacement chain of thought or teacher-authored solution.
- Do not introduce symptoms, history, tests, measurements, or visual findings absent from the input.
- Do not discard a valid step merely because its entropy is high, and do not preserve an invalid step merely because its entropy is low.
- SAFE_PREFIX_END must precede the first consequential failure.
- AUDIT_WINDOW should normally overlap ENTROPY_WINDOW, but may deviate when supported by explicit medical or logical evidence.
- If only final-answer formatting or option selection is wrong and no reasoning revision is justified, set ACTION to SKIP.

D.4 Stage 2 Continuation Prompt

The continuation input contains the original question and an unfinished assistant message with the retained prefix and an error-conditioned recovery cue. Stage 1 constructs the cue from observable medical tags, ERROR_TYPE, ERROR_SUBTYPE, and KEY_ERROR. Feedback determines repair priority; the selected trajectory's own diagnosis determines cue content. The same student continues from this prefilled message.

Prompt D.4: Stage-2 Assistant-Prefill Continuation

User message:

{original medical image(s), question, and answer options}

Assistant message:

{safe_prefix}

{Natural transition} {MINIMAL_HINT from Stage 1}

MINIMAL_HINT is inserted verbatim after the retained prefix. The student receives only the operational cue; structured error labels and the audit remain metadata. Generation continues immediately after the cue within the same assistant message, without repeating or modifying the prefix.

The SFT export retains the student-style recovery cue. Repaired and original correct examples use full-response loss

in Equation 13, supervising all assistant-response tokens. No audit or external hint is supplied at final evaluation. Record exported examples and token masks with the training configuration.

E Case Study

We present two medical image question-answering cases, including the question, options, and student model output, with the hint marked within the complete reasoning text. We preserve the wording of the student outputs without language editing. Purple labels separate the original text, hint, and continuation; blue emphasis marks selected observations and reasoning actions in the quoted outputs.

E.1 Case 08: Chest radiograph

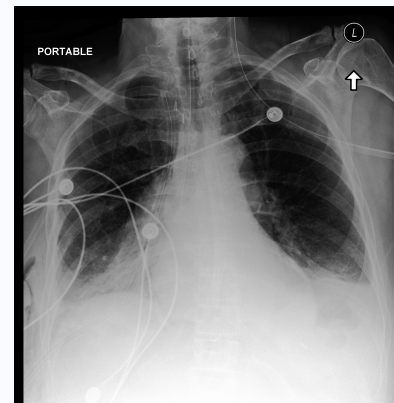
Original question and options

A 67-year-old man is recovering in the surgical intensive care unit following an esophagectomy. A portable anteroposterior chest radiograph (Figure A) is obtained to evaluate his postoperative status. The patient has a right-sided chest tube in place. Based on the radiographic findings, which of the following is the most likely cause of the bilateral lower lung opacities?

Options:

- A. Pleural effusion
- B. Pulmonary embolism
- C. Emphysema
- D. Cavitory tuberculosis
- E. Tension pneumothorax

Please reason step by step, and put your final answer within \boxed{ }.



Student model output

Original context and initial reading

<think>

The user wants a step-by-step reasoning and a final boxed answer. Let me look at the radiograph carefully. It's a **portable AP chest radiograph** of a 67-year-old man post-esophagectomy. There's a right-sided chest tube. There are ECG leads overlying the chest. There's an NG tube or feeding tube coursing down the midline (post-esophagectomy, possibly gastric pull-up). The question asks about **bilateral lower lung opacities**. Looking at the image, the lower zones do not show convincing increased opacity; the costophrenic angles are not clearly blunted, and there is no definite meniscus-like contour. The hemidiaphragms remain relatively well defined. So pleural effusion is not strongly supported.

Navigation hint

Before weighing the choices, **re-check the morphology of the basal opacities: costophrenic-angle blunting, a meniscus-like contour**, and partial obscuration of the hemidiaphragms would localize the process primarily to the...

Student continuation after the hint

Now, **let me think about the other options**: pulmonary embolism doesn't usually cause bilateral lower lobe opacities (may cause atelectasis or infarction, but not typical). Emphysema causes hyperlucency, not opacity. Cavitory TB causes upper lobe cavities, not bilateral lower lobe opacities. Tension pneumothorax causes hyperlucency and mediastinal shift, not opacity. So the answer is A, pleural effusion.

The question says "bilateral lower lung opacities" — in a postoperative patient with a chest tube, bilateral pleural effusions (or a chylothorax after esophagectomy) are common. The radiographic appearance of basal opacities with meniscus and blunting of the costophrenic angles supports pleural effusion.

Therefore the

Let me describe the radiographic findings in more detail: bilateral lower zone opacities with a meniscus sign, blunting of the costophrenic angles, obscuration of the hemidiaphragms; the chest tube is on the right; the left side shows a similar opacity. No evidence of hyperlucency or pneumothorax; no cavitory lesions; no findings of emphysema (which would be lucent). PE is not a radiographic diagnosis on chest X-ray anyway and wouldn't present as bilateral basal opacities.

Therefore the answer is A.

I should present step-by-step reasoning: identify the findings, exclude the options, and conclude with A.

Let me double-check the image description once more: the left costophrenic angle is blunted with a meniscus; the right base shows opacity consistent with effusion and possibly some atelectasis. The question asks for the "most likely cause" —

pleural effusion.

Also, after esophagectomy, patients often develop pleural effusions due to chylothorax or transudate from fluid shifts. So A is clinically plausible as well.

Format: the answer should be moderately detailed, with step-by-step reasoning, and a final answer. Use headings or bullet points sparingly. Let me write a concise structured analysis: findings, differential exclusion, conclusion. Then give the final
Let me write it like this:

Step 1 — Systematic reading: portable AP film; cardiac silhouette is difficult to assess; bilateral lower zone opacities; blunting of the costophrenic angles with a meniscus-like upper border; obscuration of the hemidiaphragms; right-sided chest tube (likely for drainage of pleural fluid/air); no evidence of pneumothorax (no visceral pleural line with hyperlucency); no cavitary lesions; lung volumes are not hyperinflated.

Step 2 — Match with options:

- Pleural effusion: fits — dependent (basal) opacities with meniscus, blunting of the costophrenic angles, common after esophagectomy (including chylothorax).
- PE: not visible on CXR; wouldn't cause diffuse basal opacities.
- Emphysema: causes hyperlucency, flattened diaphragms — opposite of opacity.
- Cavitary TB: upper lobe cavitary lesions, not bilateral basal opacities.
- Tension pneumothorax: hyperlucent hemithorax, mediastinal shift away — not present.

Conclusion: A.

I might also mention that the chest tube on the right is for drainage of pleural fluid/air, and that the left effusion is likely also present.

Let me double-check the arrow in the image: it points to the left shoulder/clavicle area — probably just a marker.
Not relevant.

Final

I'll write the response with clear reasoning. Format: short paragraphs or bullet points. Let me use a structured but concise approach</think>

Most appropriate answer: \boxed{A}.

E.2 Case 31: Elbow radiograph

Original question and options

A 32-year-old patient presents to the emergency department with acute right elbow pain, swelling, and limited range of motion following a fall onto an outstretched hand. An anteroposterior radiograph of the right elbow is obtained (Figure A). What is the most likely diagnosis?

Options:

- A. Normal elbow radiograph
- B. Isolated elbow dislocation
- C. Elbow fracture
- D. Advanced osteoarthritis
- E. Osteochondritis dissecans

Please reason step by step, and put your final answer within \boxed{ }.



Student model output

Original context and initial reading

<think>

The user wants a step-by-step reasoning for a radiograph interpretation question. **Let me carefully look at the image.** The image is labeled "R" (right) and shows an elbow radiograph. The question states it's an AP radiograph, though the image looks somewhat like an oblique or lateral-ish view, but let's focus on the pathology. Looking at the radiograph: I see the **distal humerus, the proximal radius, and proximal ulna.**

Navigation hint

Before committing, **trace the radial head and neck cortices** to verify that the horizontal lucency causes **true cortical interruption or fragment displacement** rather than projectional overlap, then **assess whether the radiocapitellar and ulnohumeral relationships** are...

Student continuation after the hint

The radiocapitellar alignment appears normal (the radial head is opposite the capitellum), and the ulnohumeral articulation is also maintained, suggesting no dislocation. In case of an isolated dislocation, the radial head would be displaced from the capitellum. Here the alignment looks preserved, so an isolated dislocation is unlikely. The presence of a fracture line at the radial head suggests a radial head fracture, so the answer is C (elbow fracture). Osteoarthritis wouldn't present acutely after trauma, and there are no osteophytes. Osteochondritis dissecans is a chronic condition with a loose body, not an acute trauma presentation.

Therefore, the most likely diagnosis is C: elbow fracture (radial head fracture). Let's describe the reasoning steps:

1. Verify patient history: 32-year-old, fall on an outstretched hand (FOOSH) -> typical mechanism for radial head/neck fractures, coronoid fractures, lateral collateral ligament injuries, etc.

2. Radiographic findings: AP view of the right elbow shows a lucent line with cortical disruption at the radial head/neck, with some displacement/fragmentation - consistent with a radial head fracture. The radiocapitellar and ulnohumeral joints remain congruent, excluding dislocation. No evidence of advanced osteoarthritis (no joint space narrowing, osteophytes, subchondral sclerosis). No chronic loose body or cartilage defect suggestive of osteochondritis dissecans.

3. Therefore the answer is C.

Also, it's worth mentioning that the "fat pad sign" may be present (elevated anterior fat pad), indicating an intra-articular fracture/hemarthrosis, supporting the diagnosis of a fracture. In the image, there may be a faint anterior fat pad sign. But the key is the radial head fracture.

Let me double-check the image description once more: the image shows the humerus, ulna, and radius. In the proximal radius region, there's a clear horizontal white line - a fracture of the radial head. The radial head may also be comminuted. So the answer is C.

Let me also consider the possibility of a coronoid fracture. Regardless, it's an elbow fracture.

Thus, the final

Format: The answer should be moderately detailed, including step-by-step reasoning: mechanism, radiographic findings, exclusion of other options, conclusion. Use headings or bullet points. Final

Let me carefully describe the image: In the image, the radial head shows a transverse lucency consistent with a fracture; the radial head fragment is slightly displaced; the radiocapitellar relationship is preserved; the ulnohumeral angle is normal; no dislocation.

Let me also mention that the lateral fat pad sign supports an intra-articular fracture.

Let me write the response.

Also, the question states "anteroposterior radiograph," but the image is somewhat lateral; I won't overemphasize the view. Final

****Step 1 – Mechanism of injury:**** A fall onto an outstretched hand (FOOSH) is the classic mechanism for a radial head/neck fracture, coronoid fracture, or lateral collateral ligament injury in a young adult.

****Step 2 – Systematic review of the radiograph:****

- ****Joint alignment:**** The radiocapitellar and ulnohumeral relationships are preserved – the radial head remains opposite the capitellum and the ulna articulates normally with the trochlea. This argues ****against an isolated elbow dislocation**** (option B), which would show obvious displacement of these relationships.

- ****Bony detail:**** There is a transverse lucent line with cortical disruption and a small displaced fragment at the radial head/neck region – i.e., a ****fracture of the radial head****. A faint anterior fat-pad sign (sail sign) is also suggestive of an intra-articular fracture/hemarthrosis.

- ****No degenerative changes:**** There is no joint-space narrowing, osteophyte formation, or subchondral sclerosis, excluding ****advanced osteoarthritis**** (option D), which would also be unlikely to present as an acute traumatic event.

- ****No chronic lesion:**** There is no well-circumscribed subchondral defect with a loose body, which would suggest ****osteochondritis dissecans**** (option E); that condition is chronic/overuse-related, not acute post-traumatic.

- The film is clearly ****not normal**** (option A) given the visible fracture line and fragment.

****Step 3 – Synthesis:**** Acute trauma + FOOSH mechanism + visible fracture at the radial head with preserved joint alignment = a fracture of the elbow (radial head fracture), not a dislocation or degenerative/chronic lesion.

****Most likely diagnosis: C. Elbow fracture**** </think>

Most appropriate answer: \boxed{C}.

F Additional Related Work

This section expands the concise main-text review with background on medical evaluation, supervision quality, localized correction, and alternative forms of self-evolution. It also covers complementary work in agent safety, robotics, and data generation.

F.1 Medical Reasoning, Distillation, and Data Selection

Medical vision-language models use biomedical instruction tuning [2, 3], generalist reasoning with **Lingshu** [11], and English–Arabic medical image understanding in **BiMediX2** [12]. Verifier-guided search, knowledge-enhanced synthesis, and reinforcement learning[36, 37, 38], together with parallel reasoning and multi-agent collaboration[39, 40, 41], improve reasoning capabilities. Process-level diagnostics and distillation audits distinguish answer correctness from reliable visual recognition, knowledge application, and evidence integration[14, 15, 42].

Medical hallucinations also expose limitations of evaluation. In controlled error-injection experiments, Farrow et al. [16] show that clinically relevant hallucinations can leave rubric scores unchanged when the errors fall outside predefined criteria. This finding motivates explicit factuality checks alongside rubric-based assessment, to assess clinically relevant errors beyond the rubric criteria.

Process supervision provides intermediate-step feedback[17], while trajectory synthesis and selection address rationale quality, difficulty, and distribution[43, 44]. **ChainPrune** [18] organizes self-generated reasoning paths into a tree and selects paths that reduce redundancy while preserving essential reasoning steps. Its distinction between token length and structural efficiency highlights the importance of assessing reasoning content when selecting trajectories. **On-policy distillation** emphasizes compatibility between student states and teacher capabilities[9]; reference-induced shortcuts, reasoning paths, and state coverage also shape supervision value[45, 46, 47]. **MedEvo** connects medical error diagnosis with student-centered supervision, using minimal teacher hints for local correction and retaining student reasoning for **self-distillation**.

F.2 Learning from Failed Trajectories and Localized Repair

Failed trajectories contain reusable reasoning as well as errors. Self-Refine and Reflexion improve subsequent attempts through self-feedback and verbal memory without weight updates [48, 49]. **Learning from Mistakes (LEMA)** constructs mistake–correction pairs, **TrajFusion** combines incorrect and correct trajectories, and Exploring Expert Failures (EEF) extracts useful actions from failed demonstrations [6, 10, 50]. Hindsight Hint Distillation (**HHD**) and FailForge guide recovery with concise hints or skills instead of full teacher solutions [1, 51].

Local recovery also depends on where guidance enters and how it becomes supervision. **Error-localization** studies, backtracking-based distillation, Test-Time Scaling via Error Localization (**TTEL**), and Selective Regenerative Decoding investigate localization, backtracking corrections, and prefix reuse [5, 19, 20, 52]. Self-Correction via Reinforcement Learning (**SCoRe**) learns self-correction through reinforcement learning; Project Aletheia and Pyli-gent derive supervision from verified backtracking traces or search trees; micro-reflective self-distillation combines micro-reflective trajectories with capability-dependent selection [21, 53, 54, 55]. EvoThink and Reflective On-Policy Self-Distillation (**ROSD**) instead use preference learning and reflective on-policy self-distillation [56, 57].

Within this trajectory-learning tradition, **MedEvo** focuses on medical-contextualized, **same-student local navigation**: preserving supported prefixes, guiding continuation with gold-blind auditing and tag-conditioned hints, and selecting supervision by student-relative learnability.

F.3 Data Evolution and Recursive Self-Improvement

Self-Taught Reasoner (**STaR**) and Reinforced Self-Training with Expectation–Maximization (ReST^{EM}) iteratively construct supervision through self-generated reasoning and outcome selection[7, 8]; EvoThink and **ROSD** further explore preference learning and reflective self-distillation[56, 57]. **Capability-dependent selection** further connects the supervision distribution to what a student can learn [21].

Recursive self-improvement examines how feedback changes subsequent learning experience [58]. In clinical interaction, **MediSkill-Evo** [22] evolves validated clinical, process, symbolic, and visual knowledge banks without fine-tuning the backbone, using process constraints to guide evidence-grounded actions. **MedEvo** instead updates model weights through supervised fine-tuning on repaired student trajectories: it diagnoses medical groups and error patterns on a fixed feedback set, then generates and repairs training-pool trajectories for the next round of **self-distillation**.

F.4 Failure Feedback beyond Medical Reasoning

Failure-aware approaches also organize recovery using structured failure feedback in agent-safety and robotics settings [59, 60, 61].

F.5 Analysis of Frontier-Model Reasoning

Luo et al. [62] characterize externalized reasoning in frontier models through token efficiency, reasoning-step types, and reasoning-tree structure, complementing outcome-based evaluation with trajectory-level analysis.

F.6 Broader Data and Environment Evolution

Difficulty-guided chart synthesis and multimodal generation workflows expand the supervision distribution [63, 64]. Data-strategy optimization, evolving environments, and autonomous medical curricula provide broader settings for recursive improvement [65, 66, 67]. These studies provide context for data evolution; MedEvo focuses on repairing student medical-reasoning trajectories and selecting subsequent supervision.

G Additional Baseline Details

This section explains the roles of the comparison models and training baselines in Table 2. The main text reports benchmark results; the descriptions below distinguish external model comparisons from the trajectory-supervision baselines used to evaluate MedEvo.

G.1 Backbones and External Comparison Models

The primary backbone is Qwen3.5-9B, with Qwen3.5-35B and Qwen3.5-397B used for the scale experiments [24]. A Base result refers to the corresponding initial checkpoint before MedEvo training. Later-round models are trained from that **same initial checkpoint** using the updated supervision dataset; the current checkpoint generates the trajectories used to construct that dataset.

The external comparison set includes general-purpose and medical-specialized models. Medical models include LLaVA-Med [2], HuatuoGPT-Vision [32], BiMediX2 [12], Lingshu [11], and HealthGPT [35]. InternVL2.5 and InternVL3 provide general-purpose multimodal comparisons [33, 34]. Table 2 groups models by proprietary access and open-source parameter scale. These model comparisons provide performance context; they are distinct from **controlled ablations** of the MedEvo training procedure.

G.2 Trajectory-Supervision Baselines

Baseline 1: HHD. Hindsight Hint Distillation uses concise hints to scaffold recovery [1]. Its role in the comparison is to assess hint-based supervision alongside MedEvo’s **medical diagnosis, prefix preservation, and feedback-driven data updates**. The cited work supplies the underlying method; the medical-benchmark scores in Table 2 are those reported in this study.

Baseline 2: TrajFusion. TrajFusion combines incorrect and correct trajectories to construct training supervision [10]. It provides a comparison for learning from failed reasoning. MedEvo instead constructs a **corrected suffix** from the same student’s retained prefix and uses medical feedback to prioritize later repairs.

Baseline 3: Teacher SFT. This baseline uses supervised fine-tuning on supervision generated by Qwen-3.8-Max. It compares **teacher-generated supervision** with MedEvo’s repaired student trajectories. We construct this baseline for the supervision-source comparison.

G.3 Reading the Comparisons

The medical icon identifies medical-specialized models, and the repair icon identifies trajectory-repair methods. Blue shading identifies MedEvo. Bold and underlining indicate the highest and second-highest distinct scores within each model group and column, with ties sharing a rank. The **seven-benchmark average** weights each benchmark equally. It is distinct from repair success, which measures successful corrections among questions submitted for repair.

Component ablations test full rewriting, suffix continuation, medical-tag feedback, and error-type feedback. Their purpose is to isolate design choices within the reported setup. Training configuration and information-access rules are provided in Appendices C.1.5 and C.1.3, respectively.